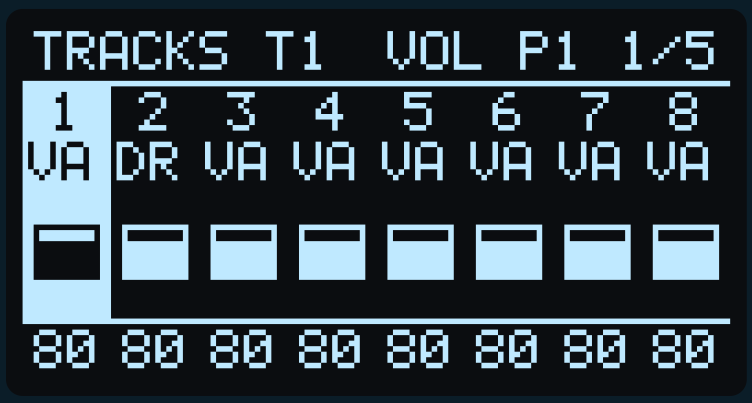
OWNER'S MANUAL

# Neptune Groovebox

An eyes-free groovebox played entirely from a Launchkey Mini 25 MK4: eight sound engines, a step sequencer with parameter locks, per-pad pressure, and sampling.




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Firmware NPE · macOS

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# 1 Welcome

Neptune Groovebox turns a Novation Launchkey Mini 25 MK4 into a self-contained instrument. The Mac makes the sound; everything you play, edit, sequence and save is done from the Launchkey, guided by its small display. You never need to look at the Mac.

The design borrows the working habits of hardware grooveboxes: a step sequencer where any step can hold its own value for any knob, a handful of modifier gestures that mean the same thing on every screen, and patterns that you chain into songs.

## What is inside

- **Eight sound engines**, one per track: a configurable analog-style synth, six-operator FM, a macro oscillator with eight models, wavetable, wave sequencer, sampler with slicing, time stretch and a granular mode, a 16-voice drum kit, and a MIDI track for outside gear.
- **Five filter models** with a live picture of the response curve.
- **Modulation**: per-voice LFOs with nine shapes and five trigger modes, extra envelopes, and pad pressure routed per track and per note.
- **Effects**: three character blocks on every track (DIRT, TALK, MODFX), insert slots with 30 effect types, two send buses, and a master EQ, compressor and limiter.
- **Sequencer**: 16 patterns, per-track length and speed, parameter locks, trig conditions, micro-timing, live recording, song mode.
- **An arpeggiator on every track**, with variation and mutation, and pressure-driven rolls on the pads.
- **Resampling**: record the master or any track into a new sample, then play it, granulate it or turn it into a wavetable.

## How to read this manual

Buttons are shown like `Shift`; names you will see on the display are shown like `CUT`. "Top pad 3" means the third pad from the left in the top row. Knobs are numbered 1 to 8 from the left.

Chapter 2 is a guided first session that ends with a short song. Chapters 3 to 12 are reference. Appendix A lists every short name on the display with its full name, where it lives and what it does.

> **STATUS** This manual describes the software as built. Much of it has so far been checked by automated tests rather than by ear and hand; where the instrument and the manual disagree, the instrument is right and the manual wants correcting.

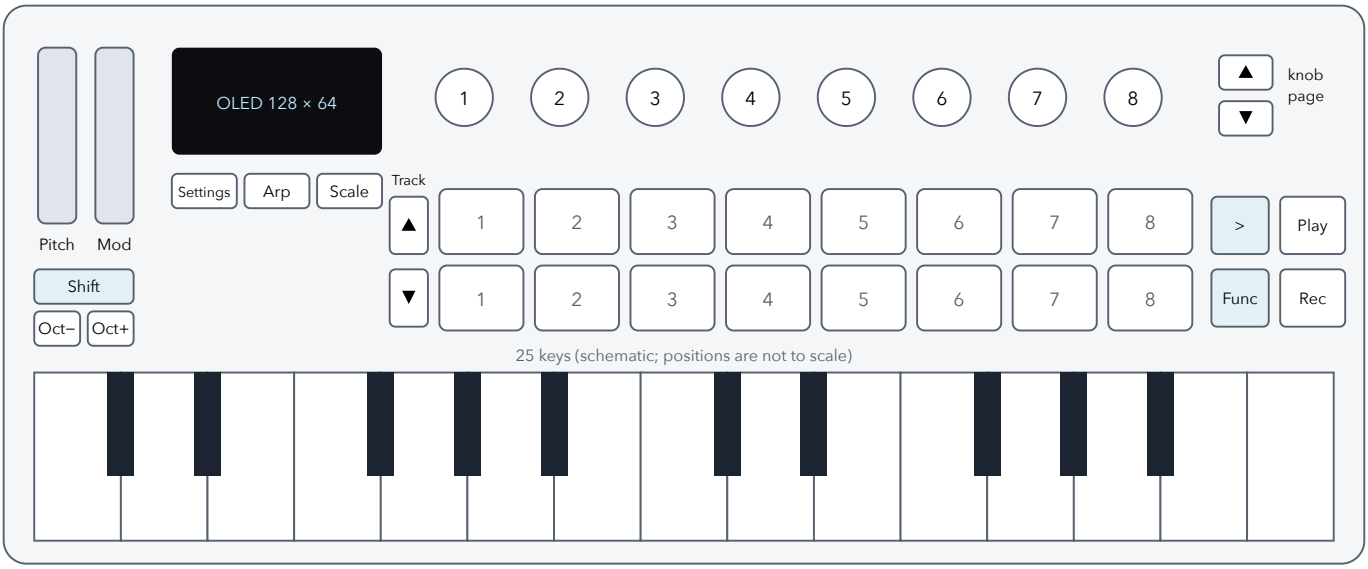
# 2 Quick start

About half an hour, from silence to a two-pattern song. Each step names the screen you should be on; if you get lost, hold `Shift` and the pads light up as a map of the screens.

> **IF YOU FORGET A COMBINATION, HOLD THE KEY** Hold `Shift`, `Func`, `>`, `▲` or `Rec` by itself for a third of a second and the display turns into a help card: what that key does with the pads, the buttons and the knobs on the screen you are on. The two small knob-page buttons turn the cards. Hold `Func` to read what copy, clear and paste would act on right now, before you press anything. Any other control makes the card step aside and does its job as usual.

## Before you start

- Plug the Launchkey into the Mac. It needs the Neptune firmware (tag NPE), which is already on your unit.
- Start *Neptune Groovebox*. The display shows the `TRACKS` screen within a couple of seconds.
- Use headphones or an audio interface if you can. The Mac's built-in speakers add a noticeable delay of their own.
- The app reopens the project you last used. For a clean start: `Settings`, press the lower knob-page button twice to reach `PROJ`, and turn `NEW` (knob 5) four clicks clockwise.




## Ten steps to a song

### 1 Look around

Hold `Shift`. The pads show where each screen lives: the bright pad is where you are. Still holding `Shift`, press top pad 2 to open `TRACKS`, the mixer. Knob 1 is the volume of track 1, knob 2 of track 2, and so on.

The two small buttons beside the knobs turn the knob page on every screen. On `TRACKS` the pages are `VOL`, `PAN`, `SND1`, `SND2` and `ENG`. The pages go round: the upper button on the first page gives the last one, and the lower button on the last page gives the first.

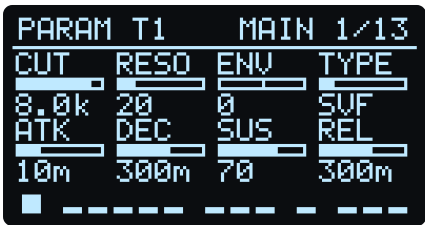
## 2 Play a sound and choose a preset

Play the keys: track 1 is a synth. Shift + top pad 5 opens PRESET. Scroll with knob 1 ( PS ) and play: each preset you land on takes over the track's sound, for listening only. Knob 2 ( CT ) narrows the list to basses, pads, leads and so on. When you like one, turn knob 5 ( LD ) four clicks clockwise to keep it. Leave without loading and your previous sound comes back.

## 3 Shape it

Shift + top pad 1 opens PARAM, the pages of the track's sound engine. It opens on MAIN: hold a note and turn CUT (knob 1) and RESO (knob 2), and a picture of the filter's response appears while you turn. Knobs 5 to 8 are the envelope ( ATK DEC SUS REL ), with a picture of its shape. The knob-page buttons step on to the engine's other pages; the full filter is on FILTR.

Three speeds work on every knob: plain, Shift held for fine, Func held for coarse jumps to useful values.




PARAM opens on MAIN




FLTR: the same filter, drawn large

## 4 Record a drum beat

Track 2 is a drum kit. Hold ▼ (the lower of the two Track buttons left of the pads) and tap top pad 2 to select track 2, then release. Shift + bottom pad 2 opens DRUMS: the sixteen pads are sixteen drum voices.

In the starting kit the bottom row holds three kicks (pads 1 to 3), a tom, a snare (pad 5), a clap, and closed and open hats (pads 7 and 8). The top row holds six tuned sub-bass hits and two cymbals.

Hold Rec and press Play: the sequencer starts from the top, recording. The name bar says REC FREE and a red light runs across the pads, one pad per step, while the pads go on playing the drums. Tap a beat: a kick on bottom pad 1, the snare on pad 5, hats on pad 7. The pattern loops after 16 steps and what you played comes round again.

Press Play to stop. Recording ends and STEP opens with the drum track selected: your beat is laid out on the pads, ready for editing. Press Play again to hear it.

Free recording keeps your timing exactly. To tighten it afterwards, raise QN on STEP (knob 3): playback is pulled toward the grid without changing what you recorded. To record straight onto the grid instead, press Rec + Play once more while recording: the name bar changes to REC QUANT.

### 5 Add a bassline with steps

Hold ▼, tap top pad 1, release: track 1 again. Open `STEP` (`Shift` + top pad 4). The sixteen pads are now the sixteen steps, top row first. Tap pads to place notes; a quick tap on a lit step removes it.

To choose a step's note, hold its pad and press a key. Do that for each step and you have a bassline.




STEP

### 6 Make one step different

This is the heart of the instrument. Go to `PARAM` and the `FILTR` page. The pads still show the steps. Hold one lit step and turn `CUT`: only that step changes, and its pad turns orange. That is a *parameter lock*. Any knob on any sound page can be locked to any step.

To remove one lock, hold the step, hold `>`, and turn the same knob.

### 7 Add space

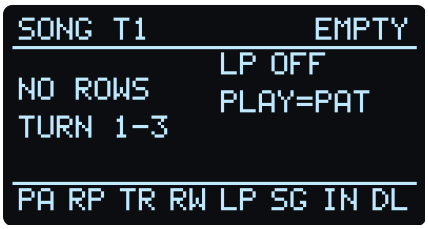
`Shift` + top pad 3 opens `TRK FX`. On its first page, `SND1` sends the track to a reverb and `SND2` to a delay. A little reverb on the snare track, a little delay on the bass.

### 8 Make a second pattern

Open `TRACKS`. Press `Func` + `Rec` to copy the whole pattern. Hold `>` and tap top pad 2 to move to pattern 2, then press `Func` + `>` to paste. Now change something in pattern 2: remove some bass steps, lock a few filter values, mute the hats by re-recording. Hold `>` and tap pad 1 or 2 to switch between the two while playing; the change waits for the end of the loop.

### 9 Chain them into a song

`Shift` + top pad 6 opens `SONG`. Turn `PA` (knob 1): row 1 appears, playing pattern 1. Set `RP` (knob 2) to 2 repeats. Tap the pad after the last row to add row 2, and set its `PA` to pattern 2. Turn `SG` (knob 6) on and `LP` (knob 5) on, then press `Play`: the song plays pattern 1 twice, pattern 2, and loops.




SONG

### 10 Save

`Settings`, then the `PROJ` page. Choose a `SLOT` with knob 1 and turn `SVAS` (knob 4) four clicks clockwise. The display answers OK. Next time the app starts it reopens this project.

## Where to go next

Each of these is a few minutes of play and opens a different side of the instrument.

**SQUEEZE** Open SCALE (Shift + bottom pad 4), hold a pad and lean into it. Then open AT ROUTE (Shift + top pad 8) and choose what pressure does on this track. Each finger has its own pressure. Chapter 7.

**PRESSURE ON ANY KNOB** On PARAM, turn a knob, then press Func + ▲. The bottom pad under that knob now sweeps it with pressure and lets go on release. Chapter 7.

**EAT YOUR OWN TAIL** Open WAVE (Shift + bottom pad 5), go to its REC page and record the master output for a bar. The take becomes a sample you can play on a sampler track. Chapter 11.

**FREEZE TIME** On a sampler track set MODE to GRAN and SCAN to 0 on the GRAIN page. The sound stands still; move STRT to walk through it by hand. Chapter 11.

**ANY LOOP, YOUR TEMPO** Put a loop on a sampler track, set MODE to LOOP, then on STRCH choose an ALGO, switch SYNC on and set LEN. Change the tempo: it follows. Try each algorithm on the same loop. Chapter 11.

**CHOP A BREAK** On SLICE set MODE to TRNS and PLAY to KEYS: every hit of the loop is now on its own key. Or PLAY = RAND with steps on every sixteenth. Chapter 11.

**A WAVETABLE FROM ANYTHING** On a wavetable track set TABL to USER, then choose a recording with SM on the WAVE screen's REC page. POS now scans through your recording, and the SCAN page can play it back at any length. Chapter 5.

**THREE WAVES, ONE TABLE** On a wavetable track open WAVE: give key 1 a sample with SM, press the second top pad and give key 2 another, then a third. Set AL to TRNS or VOC and sweep POS. Chapter 5.

**PATTERNS THAT NEVER REPEAT** Give two tracks different lengths (LN on STEP): 16 against 12, say. Then give some steps a condition (COND: PROB or A:B). Chapter 9.

**AN ARP THAT EVOLVES** Press Arp twice and hold a chord. On the VARY page raise SPCE; on MUTA switch mutation on and let it run. When you like what it has become, >PAT writes it into the pattern. Chapter 10.

**ROLLS UNDER YOUR FINGER** On the ARP screen's ROLL page switch ROLL on. Now a held drum pad repeats, faster the harder you press. Chapter 10.

**A SYNTH THAT TALKS** TRK FX, TALK page: raise MIX, sweep VOWL, add DPTH. Chapter 8.

**PUMPING** Put the DUCK effect on a bass or pad track and set its SRC to the drum track. Chapter 8.

**EVERY NOTE SLIGHTLY DIFFERENT** On the synth's ANALG page raise SLOP, FVAR and EVAR. On drums, point an LFO at TONE with TRIG = HOLD and SHPE = RND. Chapters 5 and 7.

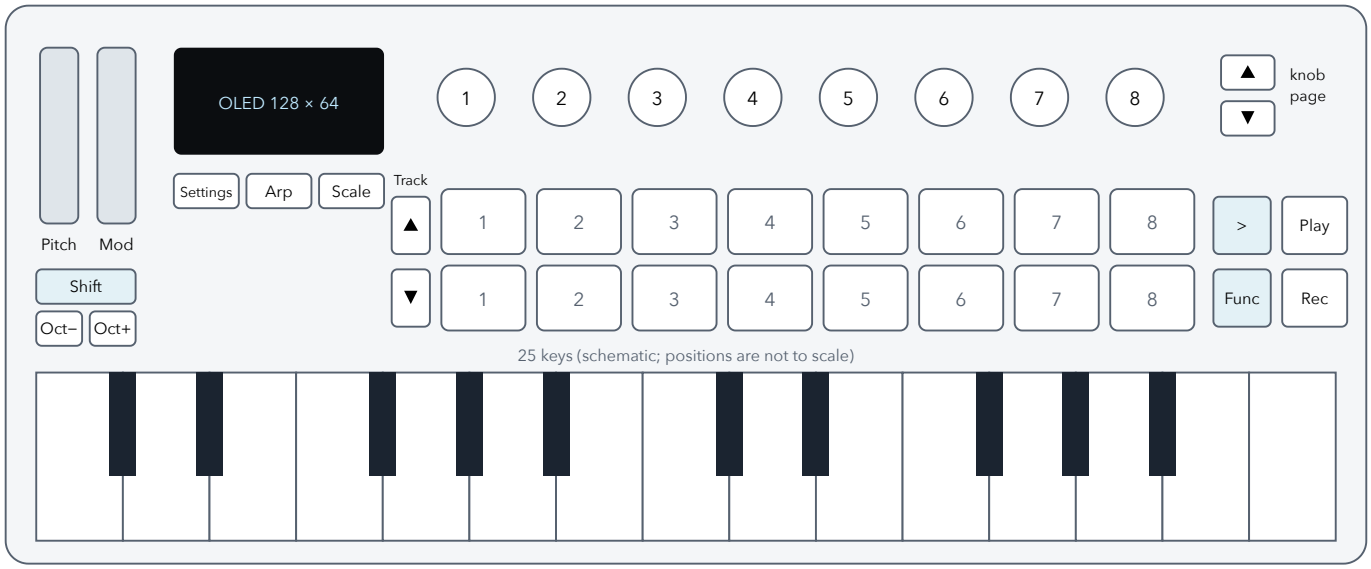
**JAM WITHOUT FEAR** Hold > and press Rec to keep a copy of the pattern. Wreck it. Hold > and press Play to bring it back. Chapter 9.

**ALL TRACKS AT ONCE** Hold ▲ or ▼ and turn a knob: the same control moves on every track that has it. Sweep every filter together. Chapter 3.

**TWO HANDS, TWO REGISTERS** `Shift` + `Oct−` drops the lowest octave of the keyboard by a further octave or more: bass in the left hand, chords in the right. Chapter 12.

# 3 Finding your way

## The control surface



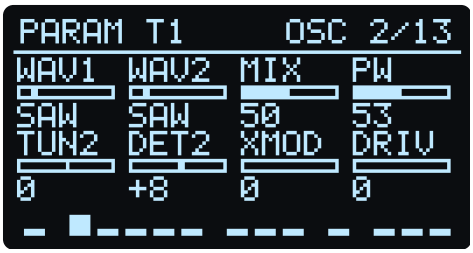

- **Eight endless knobs.** They always edit the eight things shown on the display, in the same positions.
- **Knob-page buttons** (the small pair right of the knobs): previous and next page of knobs, on every screen. The pages go round, so from the first page the upper button reaches the last in one press.
- **Sixteen pads**, velocity- and pressure-sensitive. What they do depends on the screen: play notes, hold steps, select rows.
- ▲ ▼ left of the pads (printed *Track*): tapped, they change what the pads address (the next sixteen steps, the next scale page, the next row). Held, they open the Track layer.
- `>` right of the top pad row: the pattern key.
- `Func` right of the bottom pad row: "bigger, or secondary".
- `Shift`: "where, and how finely".
- **Any of these held alone** (`Shift`, `Func`, `>`, ▲ or ▼, `Rec`) shows its help cards on the display (see *Help cards* below).
- `Settings` `Arp` `Scale` `Play` `Rec` do what they say. `Oct−` `Oct+` shift the keys.
- **Pitch strip and mod strip** act on the selected track.

## The display

The top line is the name bar: the screen or page name on the left, the selected track ( `T 1` ) on the right, and tags when something is engaged: `ARP` , `LATCH` , `PIN` , `SPLIT` .

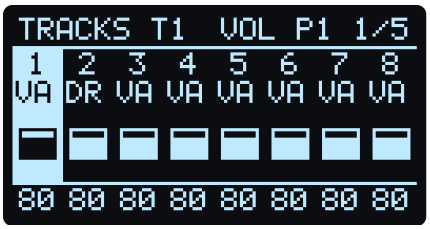
**Grid screens** show eight cells in two rows of four. Knobs 1 to 4 are the top row, 5 to 8 the bottom row.

**Bar screens** show a picture or list with a strip of eight two-letter labels along the bottom, one per knob. When you turn a knob the name bar

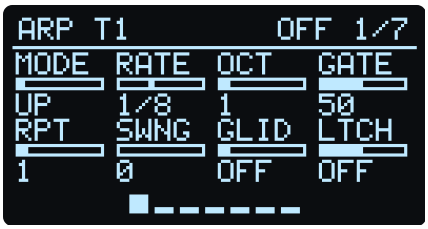



A grid screen: the synth's OSC page

spells out its full name and value. A knob that has nothing to act on at the moment says why there for a second instead of doing nothing: `SAMPLER ONLY`, `NOT ZOOMED` and `ONE SLICE` on `WAVE`, `ROUTE OFF` on `AT ROUTE`, `FLT B IS OFF`, `FILTER IS OFF` and `SEM MRPH ONLY` on the filter pages, `SLOT IS EMPTY` on `TRK FX`, `NOTHING KEPT` and `NOTHING TAPPED` on `ARP`, `SONG IS EMPTY` and `ONE ROW` on `SONG`, `NO LOCKS` on a held step.



A bar screen: TRACKS, VOL page



ARP, first page

## Screens: Shift + pad

Hold `Shift` and press a pad. The screens follow the legends printed above and below the pads.

| PAD | TOP ROW | BOTTOM ROW |
|---|---|---|
| 1 | `PARAM` – the sound engine's pages | `STEP` – the step sequencer |
| 2 | `TRACKS` – mixer and engine choice | `DRUMS` – drum pads |
| 3 | `TRK FX` – track effects | `CHORD` – chord pads (the arpeggiator is on the `Arp` button) |
| 4 | `STEP` – the step sequencer | `SCALE` – scale pads |
| 5 | `PRESET` – sound presets | `WAVE` – waveform and resampling |
| 6 | `SONG` – song rows | `FLTR` – the filter picture |
| 7 | `SETTINGS` (also the `Settings` button) | – |
| 8 | `AT ROUTE` – pad pressure routing | – |

## The grammar: one gesture, one meaning

These hold on every screen. Learn this table and you know the instrument.

| HOLD | AND TURN A KNOB | AND PRESS A PAD | AND PRESS ▲ / ▼ | AND PRESS A KNOB-PAGE BUTTON |
|---|---|---|---|---|
| nothing | edit, one step | the screen's pad action | what the pads address: step page, scale page, song row, preset list | knob page ±1 |
| `Shift` | fine: a tenth of a step (on an envelope's `ATK`, `DEC` or `REL`: that stage's curve, see Filters) | go to that screen | previous / next track | – |
| `Func` | coarse: jump to useful values | top row: jump to a page; bottom row on `TRACKS`: solo | set the low (▼) or high (▲) pressure-range end of the knob turned last | section jump, or first / last page |

| HOLD | AND TURN A KNOB | AND PRESS A PAD | AND PRESS ▲ / ▼ | AND PRESS A KNOB-PAGE BUTTON |
|---|---|---|---|---|
| a step pad | lock that knob on that step | – | – | lock page ±1 |
| `>` | on a held step: remove that knob's lock | queue pattern 1–16 | – | (`Rec` keeps a copy of the pattern, `Play` reloads it) |
| `▲` or `▼` | the same control on every track | top row: select track N; bottom row: mute track N | – | – |

### Knob speeds

Plain is 1 % of a continuous range or one item of a list. `Shift` is a tenth of that. `Func` snaps: levels to 0, 25, 50, 75, 100 % and the default; two-sided values to the ends, the centre and the quarters; tuning and frequencies by octaves; tempo by 10; lengths by 4; division lists to the straight (not dotted or triplet) values; an LFO's `DEST` list from one page's group of destinations to the next.

### Buttons

| BUTTON | PLAIN | WITH SHIFT | WITH FUNC |
|---|---|---|---|
| `Arp` | Opens `ARP`; pressed again there, switches the track's arp on or off | Latch | – |
| `Scale` | Scale lock on / off (the name bar says which, with the scale) | Opens `SCALE` | – |
| `Settings` | Opens `SETTINGS` | – | Tap tempo |
| `Play` | Start / stop; stopping also ends recording | Stop and return to the start | Clear |
| `Rec` | Recording on / off, when the button is let go. Held + `Play`: start recording; again: free / quantized | Fill, while held | Copy |
| `>` | Pattern key (hold) | – | Paste |

### Help cards

Every key that changes what the other controls do while you hold it has help cards: `Shift`, `Func`, `>`, `▲` / `▼` and `Rec`. Hold one of them by itself. After a third of a second the display becomes a help card for that key, and stays for as long as you hold it. If you used a knob, pad or button in the two seconds before, the key is taken for the start of a chord and the card waits two seconds instead, so `Shift` + turn in the middle of editing never covers the value you are setting. While a card is showing, the two knob-page buttons turn to the key's other cards (the mark at the top right, such as `2/3`, says which one you are on). Anything else you press or turn makes the card step aside for the rest of that hold and then does exactly what it always does, so a quick `Shift` + pad or `Func` + `Play` never sees a card at all. Let go and the screen is back as it was.

`Shift` has three cards, the same everywhere: the pads (a map of the screens, the one you are on inverted), the buttons, and the knobs. `Func` has three that follow the screen: *edit* (what `Rec`, `Play` and `>` would copy, clear and paste right now, and what undo and redo would take back or do again), the pads (the page keys by name), and the knobs and other buttons. `HELP` on `SETTINGS` → `SETUP` chooses when the cards appear: `OFF`, `0.35S` or `INST` (at once).

[>] has two cards: the sixteen patterns laid out like the pads (the current one inverted, the one queued next marked `*`, patterns with steps in them by number, empty ones as a dot), and what [Rec], [Play] and a knob do under it, including whether [Play] would reload a kept copy or the pattern as the project had it. [▲] / [▼] have two: the eight tracks over the top pads with their engines and, under each, `ON`, `MUTE` or `SOLO`; and what a knob does under them here and what a quick press of the buttons moves on this screen. [Rec] has one: what letting go will do, and the chord with [Play].

Two things to know. A quick press of [▲] or [▼] still pages as always; a press held long enough to show the card does not page when you let go, because the card took its place. [Rec] is the exception: it switches recording on or off when you let go whether or not its card was up. [Play], [Settings], [Arp] and [Scale] act the moment you press them and have no cards; what they do with [Shift] or [Func] is on those keys' cards.

### Copy, clear, paste

[Func] + [Rec] copies, [Func] + [Play] clears, [Func] + [>] pastes. What they act on follows where you are: a held step if you are holding one (`STEP`: the whole step; `PARAM`, `TRK FX`, `FLTR`: its locks); otherwise the track's steps on `STEP`, the whole pattern on `TRACKS`, the track's sound on `PARAM` and `FLTR`, and the track's effects on `TRK FX`. On `PARAM` and `FLTR` copy and paste take the whole sound (engine, mixer, effects), and clear resets the engine's own parameters and leaves the mixer and the effects alone, so the display calls it `RESET`. On `TRK FX` all three act on the effects only: the `DIRT`, `TALK` and `MODFX` blocks, the insert slots and the `ARPLK` values. Clearing switches the blocks off, empties the slots and sets the arp locks to off; volume, pan, sends, length and speed on the `TRACK` page are mixing and sequencing and are never touched. On other screens the three do nothing, and the name bar says so (`NO CLEAR HERE`). The name bar confirms what was done (`CLEAR T3 STEPS`, `CLEAR T3 FX`). If you are not sure what a combination will act on, hold [Func] and read the card first.

**A whole pattern needs a held button.** On `TRACKS`, clear and paste replace every track's steps, so [Func] + [Play] and [Func] + [>] act only after the button has been held for half a second. The name bar (or the card, if it is showing) fills up while you hold: `CLEAR >>>..`. Let go early and nothing happens; the name bar says `HOLD TO CLEAR`. Everything smaller acts at the press.

### Undo and redo

Hold [Func] and [Shift] and press [Play] to undo; hold them and press [Rec] to redo. The last fifteen actions can be taken back, one press each, and it does not matter what you played, turned or looked at in between: an action stays undoable until it is undone or a sixteenth pushes it out. The name bar says what was taken back (`UNDO CLEAR T3 STEPS`), and the [Func] card shows what the next undo and redo would be before you press (`UNDO: CLEAR T3 STEPS`, `REDO: ...`, or `NOTHING TO UNDO`).

These are undo steps: every clear, paste and reset; reloading a pattern with [>] + [Play]; changing an insert's `TYPE`; `LCKS` clearing a step's locks; `EVOL` turned back to the ancestor; `>PAT`; and a recording, one step from the moment it starts to the moment it stops, so a bad take is one press away from gone (also while it is still running: the take so far goes and recording carries on). Knob turns are **not** undo steps. Undoing a change to a sound puts back what that action changed and leaves knobs you turned since as they are. Loading a project or starting a new one empties the list, and a step whose track has been given another engine since is skipped.

The older shortcut still works: the same copy, clear or paste combination pressed again straight away, with nothing in between, takes it back.

### Things that destroy work

Loading, saving over, deleting and the like sit on a knob and need four clicks clockwise, shown as `>>>` filling up. Any other knob, or a click back, cancels. Nothing destructive is ever on a pad. Clearing or pasting over a whole pattern needs its button held for half a second, and whatever the edit combinations do can be undone.

### Page keys

On screens with several pages, `Func` + a top-row pad jumps straight to a page (on `PARAM`, to a section; the same pad again steps within it).

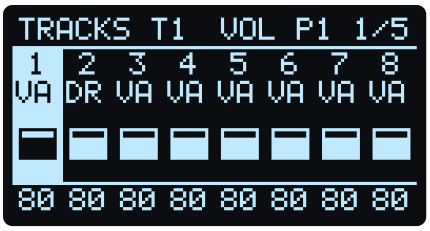
### Pinning the playing pads

On `DRUMS`, `CHORD` or `SCALE`, hold `Shift` and hold that screen's own bottom pad for half a second. `PIN` appears in the name bar and those pads keep playing, with pressure, on every other screen, so you can play pads with one hand and turn sound knobs with the other. The same gesture unpins.

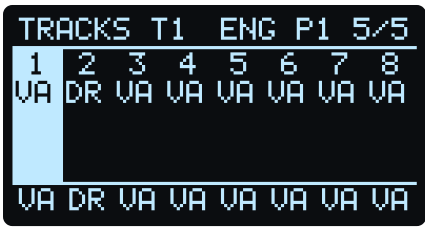
# 4 The screens

Every screen, what its knobs show and what its pads do. Short names are explained in Appendix A.

## TRACKS – the mixer




VOL



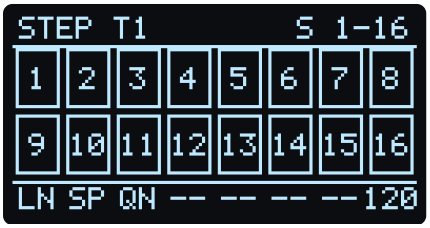

ENG

**Knobs:** knob N belongs to track N. Five pages: `VOL`, `PAN`, `SND1`, `SND2`, `ENG`.

**Pads:** the top row selects a track. The bottom row mutes (green = sounding, red = muted, blue = solo); `Func` + bottom pad solos.

**Changing a track's engine:** on the `ENG` page turn the track's knob to choose an engine. It is shown boxed as pending and nothing changes yet. Tap that track's top pad (lit amber) to commit. Leaving the page drops the choice. Changing engine replaces the track's sound in every pattern.

## STEP – the step sequencer




STEP

**Knobs:** `LN` track length, `SP` track speed, `QN` playback quantize, and the tempo on knob 8.

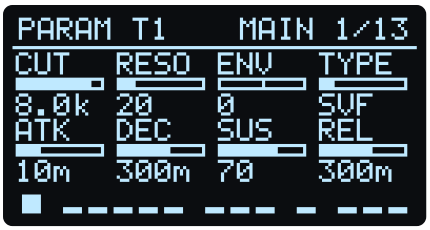
**Pads:** sixteen steps. A tap adds a step; a quick tap on a lit step removes it; a held step is never removed. `▲` / `▼` move to the next sixteen steps of a longer track. Hold a step and press a key to set its note (several keys for a chord).

**Step detail:** hold a step (a little longer than a tap) and the screen changes to that step's own pages, without anything else being touched. It opens on the track's `MAIN` page (a drum track: the `DR` page of the drum that step plays; a MIDI track: its first page), and every cell shows the value that step will play: a small mark in the cell's corner means the step has its own value there (a lock), otherwise it is the track's value. Every turn is a lock. The knob-page buttons move through all the track's sound pages while the step is held, and one press back from `MAIN` is `TRIG` (`NOTE VEL LEN MICR` | `COND PROB A B LCKS`), the step itself. The name bar shows the step (`S5 LOCK`). Let go and `STEP` is back. Turning a knob straight after pressing opens the pages at once; a quick tap never shows them. With several steps held, the pages show the one nearest the top left and every held step is edited together: a knob gives all of them the value you see, and a key gives all of them its note.

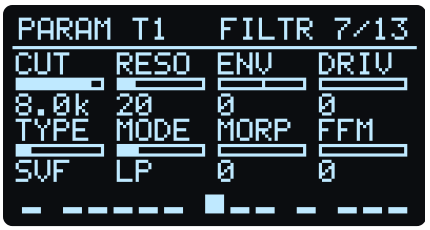
**Recorded pressure:** a step whose notes were recorded from pads with pressure has a small dark dot at the lower right of its cell, and one more page, `PRES`, one press back from `TRIG` (the pages go round, so it is also the last page). It shows the recorded curves over the note's length and three cells: `PRES` sets how much of the pressure is

played (0 to 200 %; 0 plays none and keeps the recording), `>LCK` turns it into parameter locks, `CLR` removes it. `>LCK` and `CLR` need four clicks clockwise.

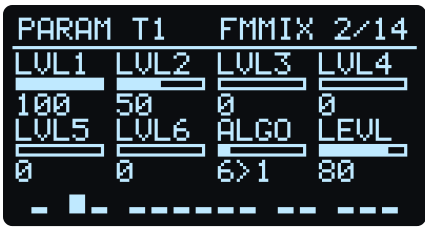
## PARAM – the sound



Synth, MAIN (the landing page)



Synth, FILTR

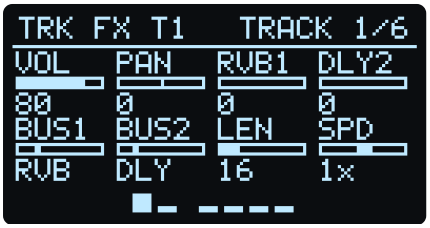


FM, FMMIX

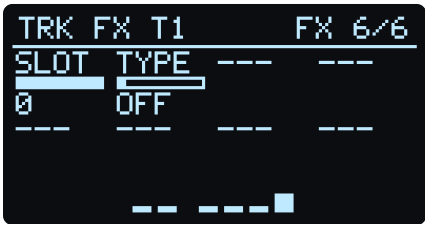
**Knobs:** the pages of the selected track's engine (Chapter 5), opening on `MAIN`: cutoff, resonance and the level envelope, the same on every engine that has a filter. A drum track shows one `DR` page for the pad last selected on `DRUMS` (and its `KICK` page when that pad is a kick machine), then its LFOs.

**Pads:** the track's steps, for locks: dim = empty, green = has a note, orange = has locks, white = held. A bottom-row pad under a knob that has a pressure range is violet and plays that range instead.

## TRK FX – track effects



TRACK



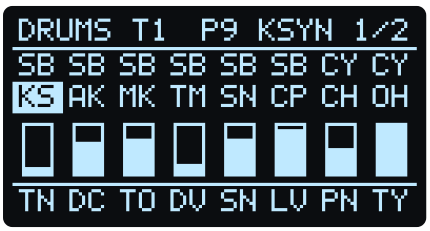
FX

The same six pages on every track, each with a page key (Func + top pad 1 to 6):

- `TRACK`: `VOL PAN SND1 SND2` | `BUS1 BUS2 LEN SPD`. The send cells are labelled by what the bus carries (`RVB1`, `DLY2`).
- `ARPLK`: `GATE VEL RTCH`, there to be locked per step so the arp changes along the pattern.
- `DIRT`, `TALK`, `MODFX`: the three character blocks (Chapter 8).
- `FX`: `SLOT TYPE ON WET` and the chosen insert's four controls.

**Pads:** as on `PARAM`; every cell here can be locked and given a pressure range.

## DRUMS – drum pads

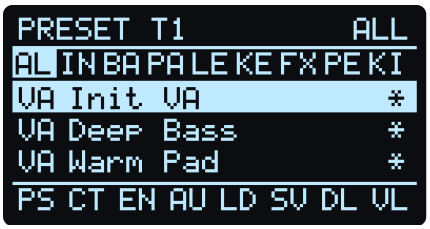


DRUMS

**Pads:** sixteen drum voices, played with velocity and pressure. One colour per kind of voice; white while it sounds. Playing a pad selects it.

**Knobs:** the selected pad's voice: `TN DC TO DV SN LV PN TY` (tune, decay, tone, drive, snap, level, pan, type). The screen follows the selected track if it is a drum track, otherwise the first drum track.

## PRESET – sound presets




PRESET

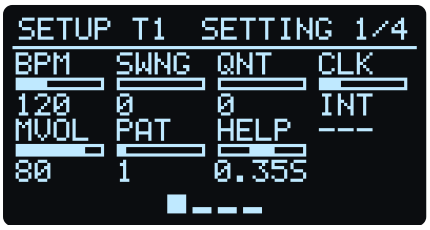
**Knobs:** `PS` scroll, `CT` category, `EN` this engine or any, `AU` audition, `LD` load, `SV` save, `DL` delete, `VL` track volume. Load, save and delete need the four clicks.

**Audition:** opening the screen changes nothing. As soon as the highlight moves (knob 1, ▲ / ▼, or a change of category or engine filter), the highlighted preset takes over the track's sound for listening only, and the name bar shows AUD. Each preset is heard in a fresh copy of its engine, exactly as it will sound once loaded (notes still sounding from the previous one end). The track's own sound comes back, untouched, when you leave the screen by any route or select another track. `LD` keeps the highlighted preset instead (the name bar shows LOAD OK): after that nothing is put back. `AU` is the off switch: turned off (AU- in the name bar), the list scrolls without changing the sound, and `LD` still loads. A project saved while you are listening stores the track's own sound.

**Pads:** play the track on the notes of the scale chosen on `SCALE` (its page and root; sixteen semitones up from middle C until you choose one, with the root lit), so you can audition without reaching for the keys. ▲ / ▼ also scroll the list.

Your own presets are files in *Music / Neptune / Presets*.

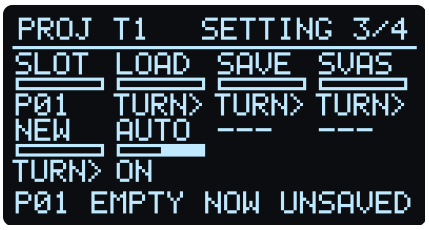
## SETTINGS



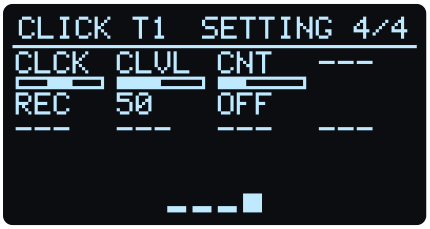

SETUP




MASTR




PROJ




CLICK

Four pages. `SETUP` : tempo, swing, quantize, clock source, master volume, pattern, and `HELP` (when the help cards appear). `MASTR` : master EQ, compressor and limiter. `PROJ` : project slots (Chapter 12). `CLICK` : the metronome –

`CLCK` (when it clicks), `CLVL` (how loud) and `CNT` (the count-in before a recording); see Recording in the sequencer chapter. The metronome's settings belong to the instrument, not to a project: loading or starting a project leaves them as they are, and they are kept between sessions.

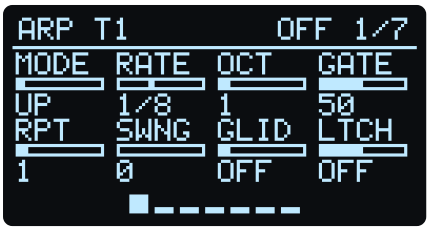
## FLTR – the filter picture



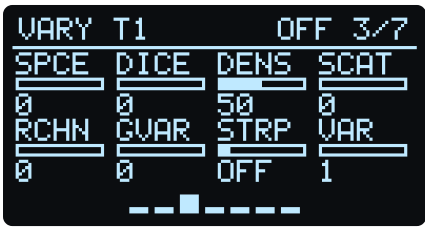

FLTR

The filter's actual response, measured from the filter itself, with the engine's filter controls beneath as two-letter labels. For the synth's two filters it shows them combined as routed. Further filter pages of the engine follow on the knob-page buttons. On the filter envelope's page, turning an envelope knob shows the envelope's shape in place of the response for a second. Drums and MIDI tracks have no filter.

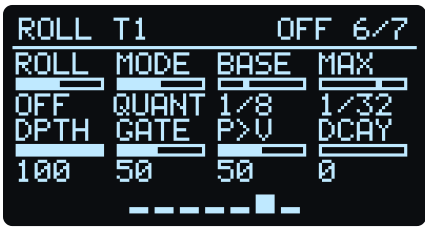
## ARP – arpeggiator and rolls




ARP



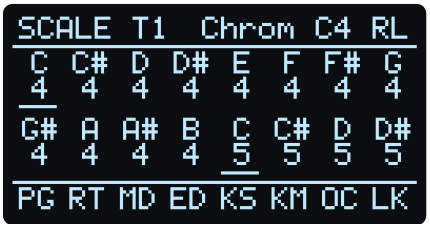

VARY




ROLL

Seven pages for the selected track's arpeggiator: `ARP`, `PLAY`, `VARY`, `MUTA`, `ACFG`, and `ROLL` / `ROL2` for pad rolls (Chapter 10). An `ARP` tag in the name bar shows the arp is on.

## SCALE – scale pads

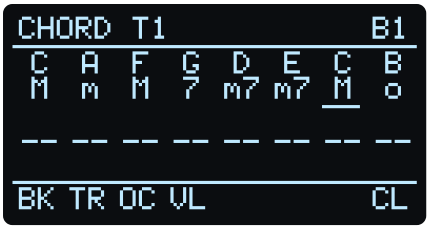



SCALE

**Pads:** play the selected track in a scale, with pressure; the root is blue, other notes green. ▲ / ▼ change the scale page. Twelve pages are built in (chromatic, major, minor, pentatonics, blues, whole-tone and modes) and you can make your own.

**Knobs:** `PG` page, `RT` root, `MD` relative or absolute, `ED` edit, `KS` and `KM` let the keys choose pages, `OC` octave, `LK` scale lock (the same switch as the `Scale` button: with the lock on, the keys play the nearest note of the page's scale on synth tracks, arps stay in it, and `SCALE` shows in the name bar). In edit mode tap a pad and press a key to assign its note.

## CHORD – chord pads




CHORD

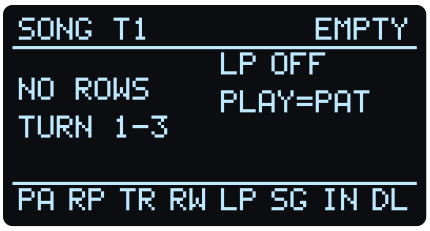
**Storing:** hold a chord on the keys and press a pad. The chord is on that pad: exactly the notes you held, each at the level you played it. Whatever the pad had before is replaced (undo brings it back). Keys kept by the latch count as held, so you can latch a chord and use the same hand for the pad. While keys are down the name bar names the chord ( `Cm7 > PAD` ) and the empty pads light up; afterwards it says where the chord went ( `PAD 5 = Cm7` ).

**Playing:** press a pad with no key held. It plays its chord on the selected track; the harder you hit, the louder the chord, and pressure reaches every note. Chord pads record into steps, feed the arp when it is on, and roll, like every other playing pad. They can be pinned (below), so a progression stays under your fingers while you edit the sound.

**Display:** the sixteen pads as two rows of eight, each with the chord's root over its kind: `M` major, `m` minor, `7` , `M7` , `m7` , `o` diminished, `h7` half-diminished, `+` augmented, `s2` / `s4` suspended, `6` , `9` , `5` a bare fifth, `*` anything else (named by its lowest note). A line under a cell marks an inversion; `--` is an empty pad. The pad lights are coloured by root, neighbours on the circle of fifths in neighbouring colours, so a progression reads as colours.

**Knobs:** `BK` bank (four banks of sixteen; also ▲ / ▼), `TR` transpose and `OC` octave for every pad, `VL` whether each note keeps the level it was stored with, `CL` clear the bank (four clicks). Chords are saved with the project.

## SONG

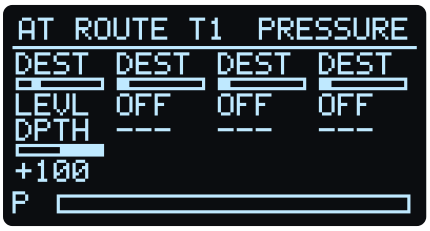



SONG

**Knobs:** `PA` pattern, `RP` repeats, `TR` transpose, `RW` row, `LP` loop, `SG` song on / off, `IN` insert a row, `DL` delete the row (four clicks).

**Pads:** pad N selects row N; the pad after the last row adds one. With `SG` on, `Play` follows the rows; a row ends when track 1 has wrapped the set number of times.

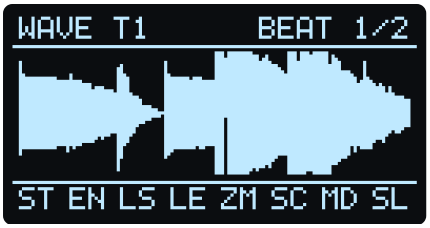
## AT ROUTE – pad pressure



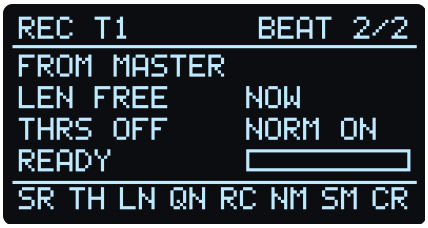

AT ROUTE

What pad pressure does on the selected track. Top row: the destination of routes 1 to 4. Bottom row: how far each moves. Every track has its own routes; the destinations on offer depend on the engine (Chapter 7). The pads play the track chromatically so you can feel the result.

## WAVE – waveform and resampling




WAVE




REC

`WAVE` page: the track's sample with start and end brackets, loop marks, slice ticks and a moving line for each sounding voice (in granular mode, a tick per grain). Knobs: `ST EN LS LE` markers, `ZM` zoom, `SC` scroll, `MD` play mode, `SL` slices. Pad N selects slice N.

On a wavetable track the page is the key-frame editor of the `KEYS` table (Chapter 5): the sample of one key, a line where its cycle is taken and a tick for every other key on the same sample. Knobs: `KY` key, `SM` its sample, `PO` its position in the sample, `PL` its place along the table, `ZM` zoom, `SC` scroll, `AL` algorithm, `TB` table. The top-row pads choose the key; the lit ones are in use.

`REC` page: resampling (Chapter 11).

# 5 Sound engines

Each track has one engine, chosen on TRACKS → ENG. Its pages appear on PARAM. All engines except drums and MIDI have a filter (Chapter 6); all sound engines have LFOs (Chapter 7).

### The landing page: MAIN

PARAM opens on the same page for every engine that has a filter, so the first handles of a sound are always under the same knobs: knob 1 is the filter cutoff ( CUT ), knob 2 the resonance ( RESO ), knob 3 how far the filter envelope moves the cutoff ( ENV ; on FM, macro, wavetable, wave sequencer and synth; the sampler shows ENVF ), knob 4 the filter TYPE , and knobs 5 to 8 are the envelope that shapes the level: ATK , DEC , SUS , REL (on FM, operator 1's). These are the same cells as on the engine's FILTR and AMP pages further on: a value, a parameter lock or a pressure range set here is set there too. Turning them shows the filter or envelope picture. The engine's own pages follow, in the tables below.

### How the engines relate

The engines are separate instruments, one per track. The synth is the one that mixes kinds of oscillator: each of its three oscillators can be an analog wave, a wavetable, an FM pair, a formant oscillator or a harmonic oscillator, in any combination, into the same mixer, filters and envelopes (see the synth, below). The FM, macro, wavetable and wave-sequencer engines go deeper into one idea each (six operators, physical models, timed scans through a recording, sequenced waves) and are chosen per track. ARCH (mono, hybrid, poly) only loads different starting values into the synth's own pages. Every engine shares what comes after the oscillators: the same five filter models, the same envelopes and LFOs, the same effects and the same sequencer.

There is also a bridge between engines: resampling. Any track (a macro oscillator, an FM sound, the synth) can be recorded on the WAVE screen's REC page, and the recording becomes a sample for a sampler track or the USER table of a wavetable track (Chapter 11). That is how a macro-oscillator sound becomes a wavetable. To layer two engines on one note, put them on two tracks and play or sequence both.

## Synth – the configurable voice

Three oscillators and noise into two filters and an amplifier, up to sixteen voices. ARCH on the VOICE page loads a starting point (mono, hybrid, poly) into the same pages; MODE sets mono, duo, poly or unison; STCK stacks detuned copies. PW shapes every waveform. At 50 each wave is plain, and the two directions are two different sounds that grow smoothly out of it:

| WAVE | BELOW 50: THINNER, SHARPER | ABOVE 50: THICKER |
|---|---|---|
| SQR PLS | The pulse narrows: classic pulse width, thin and nasal at the bottom. | A square an octave higher is mixed in, up to half and half. |
| SIN TRI | Wavefolding: the wave is driven up to six times too hard and the excess folded back, for bright, glassy harmonics. | Saturation: the wave is pushed into a limiter and squares off, warm then hollow. |
| SAW | A second saw is added, a part of a cycle behind the first: the doubled, phasing saw of saw PWM. | A saw half a cycle behind is taken away: the saw hollows out into a square. |

An LFO sent to PW animates whichever wave is selected; centred on 50 it sweeps through both sounds.

### Oscillator kinds: wavetable, FM, formant, harmonic

WAV1 , WAV2 and WAV3 go on past the analog waves to four other kinds of oscillator: WT , FM , FORM and HARM . Any slot can be any kind, so a wavetable can sit beside a saw and an FM pair, all three through the same filter. Tuning,

detune, cross-modulation, the stack and the mixer treat every kind alike. Each non-analog slot has two controls of its own on the `SRC` page, labelled with the slot's number (slot 1 on knobs 1–2, slot 2 on 3–4, slot 3 on 5–6; a slot playing an analog wave shows nothing there):

| KIND | FIRST CONTROL | SECOND CONTROL | WHAT IT IS |
|---|---|---|---|
| `WT` | `POS` position in the table | `TBL` table: SAW SQR TRI SIN PWM FORM HARM SPEC DIGI, then VOWL SYNC FOLD FM1 FM3 BELL ORGN NOIS RESO WLSH PRIM OCTV PDST COMB | The wavetable engine's oscillator in a slot: any of its 23 built-in tables, read the same way; sweep POS for movement. |
| `FM` | `IDX` modulation depth | `RAT` modulator ratio, 1–12 | A sine modulated by a second sine at a whole multiple of its pitch: bells, basses, glassy tones. |
| `FORM` | `FRM` formant frequency, 1–16 times the pitch | `WID` width of the formant | A resonant, vocal peak that stays put while the note changes character around it. |
| `HARM` | `TLT` tilt, bright to dark | `ODD` all harmonics to odd only | Up to twelve harmonics added together; odd-only is hollow like a square. |

Whatever moves `PW` (an LFO with `DEST` `PW`, pad pressure routed to `PW`) also moves the first control of every non-analog slot, so one LFO can sweep a pulse's width and a wavetable's position together. Choosing a new kind for a slot starts it from that kind's own settings. The wavetable kind is the wavetable engine's own oscillator on the wavetable engine's built-in tables (described in the Wavetable section below), read smoothly. What stays in the wavetable engine: tables made from your own recordings (`USER`, `KEYS`), the timed `SCAN`, and the coarse read-outs of `INTP`. An LFO sent to a slot's table cell moves it by eight tables at full depth.

| PAGE | KNOBS 1–4 | KNOBS 5–8 |
|---|---|---|
| `OSC` | WAV1 WAV2 MIX PW | TUN2 DET2 XMOD DRIV |
| `OSC3` | WAV3 TUN3 LVL3 NOIS | NCOL TUNE FINE BEND |
| `SRC` | | |
| `VOICE` | ARCH MODE VOX LEGA | STCK SPRD GLID GLMD |
| `ANALG` | DRFT DRAT SLOP PHAS | WSHP SOFT FVAR EVAR |
| `FILTR` | CUT RESO ENV DRIV | TYPE MODE MORP FFM |
| `FLT B` | TYPB MODB OFSB RESB | ROUT BAL DRVB TRKB |
| `F ENV` | ATK DEC SUS REL | CURV KEY VEL PMOD |
| `AMP` | ATK DEC SUS REL | CURV VEL LEVL |

Three more pages follow for modulation: `MOD` and `MOD2` (up to four LFOs) and `MENV` (up to four extra envelopes). See Chapter 7.

The `ANALG` page is the instrument's "age" control: drift, tuning slop, and per-note variation of waveform, filter and envelope. With everything at zero, successive notes are identical; raise them and no two are.

## FM — six operators

Six sine operators, eight algorithms. `FMMIX` puts all six operator levels under knobs 1 to 6 so you can play the timbre directly; `RATIO` does the same for their frequency ratios. Each operator then has its own page with ratio, level,

detune, feedback and envelope.

| PAGE | KNOBS 1–4 | KNOBS 5–8 |
|---|---|---|
| FMMIX | LVL1 LVL2 LVL3 LVL4 | LVL5 LVL6 ALGO LEVL |
| RATIO | RAT1 RAT2 RAT3 RAT4 | RAT5 RAT6 TUNE BEND |
| OP1–OP6 | RATI LEVL DETN FDBK | ATK DEC SUS REL |
| FILTR | CUT RESO ENV DRIV | TYPE MODE MORP KTRK |
| F ENV | ATK DEC SUS REL | CURV EVEL VEL WHL |
| STRIP | DST1 AMT1 DST2 AMT2 | DST3 AMT3 WVIB VRAT |
| LFO1 | RATE DPTH SHPE TRIG | SYNC DIV FADE DEST |
| LFO2 | RATE DPTH SHPE TRIG | SYNC DIV FADE DEST |

Algorithms: 6>1 · 2+3 · 3x2 · 2x3 · 1>3 · ADD · 2+4 · 4+2.

## Macro — eight models, four knobs

One oscillator that can be eight different things. MODL chooses the model, and knobs 2 to 5 are renamed to suit it.

| MODEL | WHAT IT IS | KNOBS 2–5 |
|---|---|---|
| VA | virtual analog | SAWS SPRD HARM MORP |
| FM | two-operator FM | INDX RATI DETN FDBK |
| FOLD | wavefolder | FOLD SHPE BIAS CHEB |
| FORM | formant | VOWL SHFT Q MORP |
| HARM | harmonic | TILT EVOD NUM STRC |
| PLCK | plucked string | DAMP PICK FDBK MORP |
| DUST | particle noise | DENS COLR RING MIX |
| BOW | bowed string | PRES POS DAMP SPD |

| PAGE | KNOBS 1–4 | KNOBS 5–8 |
|---|---|---|
| MACRO | MODL SAWS SPRD HARM | MORP TUNE FINE LEVL |
| AMP | ATK DEC SUS REL | CHAN STGS COMB BEND |
| FILTR | CUT RESO ENV DRIV | TYPE MODE MORP KTRK |
| F ENV | ATK DEC SUS REL | CURV EVEL VEL WHL |
| STRIP | DST1 AMT1 DST2 AMT2 | DST3 AMT3 WVIB VRAT |
| LFO1 | RATE DPTH SHPE TRIG | SYNC DIV FADE DEST |
| LFO2 | RATE DPTH SHPE TRIG | SYNC DIV FADE DEST |

## Wavetable

`TABL` picks a built-in table, `USER`, a table built from any sample (64 frames, so `POS` scans through the recording), or `KEYS`, a table you define with key frames (below). `MSPD` and `MDPT` move the position by themselves; `ENVP` and `VELP` move it with the envelope and velocity. `INTP` sets how the table is read, from smooth to deliberately coarse. Its last setting, `CLSC`, reads any table the way the first digital wavetable synths did: the scan moves in 64 whole steps, the wave is 128 samples of 8 bits with nothing smoothed in between, and nothing is band-limited to the note, so high notes get a glassy top of tones that are not harmonics. (`STEP` is stepped but clean; `VINT` softens each step.)

After the first nine tables come fourteen more: `VOWL` the vowels A E I O U, `SYNC` an oscillator-sync sweep, `FOLD` a sine driven into a wavefolder, `FM1` and `FM3` FM with a rising index (modulator at 1× and 3× the pitch), `BELL` bar-like partials from a soft to a hard strike, `ORGN` an organ pulled out drawbar by drawbar, `NOIS` noise settling into a tone, `RESO` a resonant filter sweep, `WLSH` square-cornered digital waves, `PRIM` prime-numbered harmonics, `OCTV` octaves only, `PDST` phase distortion from sine to saw, and `COMB` a saw against a sliding copy of itself. All are level-matched and free of aliasing in themselves.

Each of the fourteen has at least one factory preset on `PRESET` (Vowel Lead, Sync Sweep Lead, Fold Bass, FM Table Keys, Glass Pluck, Struck Bar, Drawbar Organ, Noise To Tone, Reso Sweep Bass, Walsh Stepper, Prime Shimmer, Octave Climb, Phase Bass, Comb Phase Pad and others), and three play a table through `CLSC`: Eight Bit Vowels, Glassy Bell and Raw Sync Lead. The same 23 built-in tables can be played by the synth's oscillator slots (kind `WT`).

The `SCAN` page moves the position by itself over a set time, whatever note you play: this is time stretch for a wavetable. `MODE` runs once and holds (`RUN`), loops, goes there and back (`PING`), moves in audible steps (`STEP`) or jumps to random frames (`RAND`). The pass lasts `TIME` seconds, or with `SYNC` on, `LEN` at the tempo; with `SYNC` = `NAT` a USER table plays back at the original recording's length, and `SPD` then slows it or speeds it without changing pitch. The scan starts from `POS`.

`FRMS` chooses how a recording is cut into the USER table. The `A64`, `A128` and `A256` settings align each frame to the sound's pitch period: scanning is far smoother and the table plays in tune with the keys. They work on pitched material; noise and silence are cut the plain way.

### Key frames: your own table

The `KEYS` table is made from up to eight single waves, the key frames, each picked out of a sample. You say where each one sits along the table, and Neptune computes the frames in between.

1. On the wavetable track open `WAVE` (`Shift` + bottom pad 5). On this track its first page is the key editor. `KY`, or a top-row pad, chooses one of the eight keys.
2. `SM` gives the key a sample: a factory sample, a file, or a recording made on the `REC` page (a take made while the table is `KEYS` goes to the chosen key). The table switches to `KEYS` and you hear the key at once.
3. `PO` moves the line on the waveform: one cycle is cut there. Where the sample has a pitch the cycle is exactly one period, so the wave plays in tune with the keys whatever the pitch of the recording; in noise or a drum hit a short piece is taken. `ZM` and `SC` zoom and scroll for a finer aim.
4. Do the same for a second and a third key, from the same sample or from others. `PL` sets where each sits along the table; only their order and spacing count, the earliest key is the start of the table and the latest the end.
5. `AL` chooses how the frames between two keys are made. Sweep `POS` (or use the `SCAN` page) and listen.

The same settings are cells on four pages of the track: `K SMP`, `K POS` and `K PLC` hold the sample, position and place of the eight keys, one knob per key, and `KEYS` holds the rest.

| `ALGO` BETWEEN TWO KEYS | |
|---|---|
| `LIN` | A plain crossfade. Harmonics the two keys hold in opposite phase pass through silence. |

| ALGO | BETWEEN TWO KEYS |
|---|---|
| STEP | Nothing in between: each key holds until the next takes over. |
| CLSC | A crossfade stored as an early digital instrument stored waves: 128 samples of 8 bits. With INTP on CLSC and a low KFRM this is the whole classic character. |
| PHAS | Each harmonic's level and phase move separately, so nothing cancels on the way. |
| TRNS | Optimal transport: the energy of the wave glides along the harmonics. A formant or a single partial slides from where it is in one key to where it is in the next, instead of fading out in one place and in at another. |
| VOC | Vocoder style: first the outline of the spectrum (its formants) changes, then the harmonics under it. Half way you hear one key's formants on the other key's harmonics. |

KFRM sets the number of frames (8 to 256), BIAS leans the in-between frames towards the earlier or the later key, and WRAP brings the table back to its first key at the end, for a looped scan without a jump. ALGN lines the keys up in phase (leave it on unless you want the raw cuts), NORM evens out the level of the frames, TILT darkens or brightens every key and HARM removes the harmonics above a number. Each change rebuilds the table, so these cells cannot be locked to steps. Key frames are saved with the project; their samples are found again by name.

| PAGE | KNOBS 1–4 | KNOBS 5–8 |
|---|---|---|
| WAVE | TABL POS MSPD MDPT | ENVP VELP INTP LEVL |
| SCAN | MODE TIME SYNC LEN | STEP DIR SPD FRMS |
| OSC | OCT TUNE FINE BEND | GLID VIBR SPRD SUB |
| AMP | ATK DEC SUS REL | VEL CHAN STGS COMB |
| KEYS | ALGO KFRM BIAS WRAP | ALGN NORM TILT HARM |
| K SMP | SMP1 SMP2 SMP3 SMP4 | SMP5 SMP6 SMP7 SMP8 |
| K POS | POS1 POS2 POS3 POS4 | POS5 POS6 POS7 POS8 |
| K PLC | PLC1 PLC2 PLC3 PLC4 | PLC5 PLC6 PLC7 PLC8 |
| FILTR | CUT RESO ENV DRIV | TYPE MODE MORP KTRK |
| F ENV | ATK DEC SUS REL | CURV EVEL VEL WHL |
| STRIP | DST1 AMT1 DST2 AMT2 | DST3 AMT3 WVIB VRAT |
| LFO1 | RATE DPTH SHPE TRIG | SYNC DIV FADE DEST |
| LFO2 | RATE DPTH SHPE TRIG | SYNC DIV FADE DEST |

Tables: SAW · SQR · TRI · SIN · PWM · FORM · HARM · SPEC · DIGI · USER · KEYS · VOWL · SYNC · FOLD · FM1 · FM3 · BELL · ORGN · NOIS · RESO · WLSH · PRIM · OCTV · PDST · COMB.

## Wave sequencer

Each note steps through a short sequence of waves. Pitch, shape, level and step time each have their own lane of up to sixteen steps, and the lanes can be different lengths ( TLEN PLEN SLEN LLEN ), so the combination drifts before repeating. On the lane pages, knob N is step N.

| PAGE | KNOBS 1–4 | KNOBS 5–8 |
|---|---|---|
| SEQ | LEN XFAD LOOP RATE | PRNG TUNE FINE LEVL |
| LANES | TLEN PLEN SLEN LLEN | SYNC SLID SHP* BEND |
| PIT A | P1 P2 P3 P4 | P5 P6 P7 P8 |
| PIT B | P9 P10 P11 P12 | P13 P14 P15 P16 |
| SHP A | S1 S2 S3 S4 | S5 S6 S7 S8 |
| SHP B | S9 S10 S11 S12 | S13 S14 S15 S16 |
| LVL A | L1 L2 L3 L4 | L5 L6 L7 L8 |
| LVL B | L9 L10 L11 L12 | L13 L14 L15 L16 |
| TIM A | T1 T2 T3 T4 | T5 T6 T7 T8 |
| TIM B | T9 T10 T11 T12 | T13 T14 T15 T16 |
| AMP | ATK DEC SUS REL | VEL CHAN STGS COMB |
| FILTR | CUT RESO ENV DRIV | TYPE MODE MORP KTRK |
| F ENV | ATK DEC SUS REL | CURV EVEL VEL WHL |
| STRIP | DST1 AMT1 DST2 AMT2 | DST3 AMT3 WVIB VRAT |
| LFO1 | RATE DPTH SHPE TRIG | SYNC DIV FADE DEST |
| LFO2 | RATE DPTH SHPE TRIG | SYNC DIV FADE DEST |

## Sampler

Plays a sample from the bank: the built-in loop and hits, WAV or AIFF files you put in *Music / Neptune / Samples*, and anything you resample. `MODE` sets how it plays; `GRAN` switches to granular playback, with its own page. The `SLICE` page cuts the sample into slices, evenly or at its hits, and the `STRCH` page changes its speed without changing its pitch (all in Chapter 11). Edit the markers by eye on the `WAVE` screen.

| PAGE | KNOBS 1–4 | KNOBS 5–8 |
|---|---|---|
| SAMPL | SMPL MODE STRT END | LPST LPEN ROOT NORM |
| GRAIN | SIZE DENS SCAN SPRY | PJIT SPRD SHAP RVRS |
| SLICE | MODE SLCS SENS GAP | SLIC PLAY FADE SLEN |
| STRCH | ALGO SPD SYNC LEN | SIZE XFAD JIT SPRD |
| TONE | PTCH FINE BITS RATE | DRIV LEVL KEY BEND |
| FILTR | CUT RESO TYPE MODE | VELF ENVF KTRK FDRV |
| AMP | ATK DEC SUS REL | VEL VOX PAN RAND |
| STRIP | DST1 AMT1 DST2 AMT2 | DST3 AMT3 WVIB VRAT |
| LFO1 | RATE DPTH SHPE TRIG | SYNC DIV FADE DEST |
| LFO2 | RATE DPTH SHPE TRIG | SYNC DIV FADE DEST |

## Drums

Sixteen voices, each a small synthesizer with eight controls. Play and select them on `DRUMS`; edit the selected one there or on `PARAM`. Closed and open hats cut each other off. The hats are six square oscillators at unrelated pitches through band-pass filters, the classic drum-machine recipe; `TUNE` moves the whole cluster and `TONE` the high-pass. The open hat is open for as long as the pad, key or step lasts and closes when it ends: tap it for a closed hat, hold it (or give its step a longer `LEN`) for an open one.

| PAGE | KNOBS 1–4 | KNOBS 5–8 |
|---|---|---|
| `DR` | TUNE DEC TONE DRIV | SNAP LEVL PAN TYPE |
| `KICK` | SWEP SWPT SCRV SHAP | KNCK KTON HOLD ACRV |
| `STRIP` | DST1 AMT1 DST2 AMT2 | DST3 AMT3 WVIB VRAT |
| `LFO1` | RATE DPTH SHPE TRIG | SYNC DIV FADE DEST |
| `LFO2` | RATE DPTH SHPE TRIG | SYNC DIV FADE DEST |

Types: `FMKK` · `KICK` · `MKIK` · `SNAR` · `FMSN` · `CLAP` · `HAT` · `OHAT` · `CYMB` · `TOM` · `RIM` · `SHKR` · `CLAV` · `SUB` · `KSYN` (FM kick, analog kick, modelled kick, noise snare, FM snare, clap, closed hat, open hat, cymbal, tom, rim, shaker, clave, sub-bass, kick machine).

The five sound cells act on every type. `DEC` is the length: a drum is a one-shot and rings for its `DEC` however short the pad press or the step (`SUB`, the bass note, is the exception: it sounds while held and fades over `DEC` afterwards). `TONE` is brightness in the way of each drum (Appendix A lists them). `DRIV` is saturation. `SNAP` is the front edge: 50 leaves the drum as it is, lower softens the onset, higher lifts it and gives the tonal drums a click. Pad pressure (`AT ROUTE`) reaches `DEC`, `TUNE`, `TONE`, `LEVL`, `DRIV` and `PAN` of the hit that is sounding; `SNAP` belongs to the strike, so pressure cannot change it.

### The kick machine (KSYN)

The lower left pad of a new kit is a kick machine; set any other pad's `TYPE` to `KSYN` and it becomes one too. A kick machine is a full kick synthesizer with a second page, `KICK`: on `DRUMS` press a knob-page button (the labels become `SW ST SC SH KN KT HD AC`); on `PARAM` it is the page after the pad's own.

- On the pad's first page: `TUNE` is the pitch the drum settles on, `DEC` how long the body rings, `TONE` a low-pass over the whole drum, `DRIV` saturation, `SNAP` how hard the front edge is.
- `SWEP`: how far above `TUNE` the pitch starts, in semitones (0 = no sweep, 72 = six octaves). `SWPT`: how long the fall takes, 2 ms to 1 s. `SCRV`: where in that time the fall happens (negative: nearly all at the start, the classic kick; positive: late).
- `SHAP`: the body's waveform, from a sine through a triangle to a squared-off wave.
- `KNCK`: the level of the knock, a short tick at the very start. `KTON`: its tone, from a dull thud that lasts about 12 ms to a bright tick of about 1.5 ms.
- `HOLD`: time at full level before the decay starts, up to 0.3 s, for long, flat 808-style bodies. `ACRV`: the shape of the decay.

Every one of these can be locked per step and given a pressure range, like any other cell. The other fourteen types are unchanged and ignore the `KICK` page.

### Playing the kit

The pads play the kit on `DRUMS` (`Shift` + bottom pad 2); on the other screens the pads do that screen's own job, unless `DRUMS` is pinned. The keys play the kit on every screen while a drum track is selected. From the lowest C, eight

keys (C to G) are the bottom pad row from left to right, so the lowest C is the lower left pad; the next eight (G♯ to D♯) are the top row; the last eight (E to B) are the bottom row once more, and the top C is the lower left pad again. Harder keys are louder. The layout repeats every two octaves, so after one press of an octave button the same rows sit an octave along, and after two the keys are as they were.

> **SEQUENCING DRUMS** Record drum patterns by playing the pads or the keys with Rec on. On STEP you enter one drum at a time: the pads show the hits of the pad last selected on DRUMS. A tap adds that drum to a step or takes it away, and the other drums on the step stay, so any number of the sixteen can hit together. A pad glows dimly, and the step number is underlined, where only other drums hit. Hold a step and press a key to add that key's drum. On the step's TRIG page NOTE and VEL are the selected drum's own; length, timing, condition and locks belong to the step.

## MIDI track

Sends notes, pitch bend, per-note pressure, program changes and controllers to outside gear. Its pages send standard controllers, so the same knobs, locks and pressure ranges work on a hardware synth.

| PAGE | KNOBS 1–4 | KNOBS 5–8 |
|---|---|---|
| MIDI | CHAN PROG VOL PAN | MOD EXPR CUT RESO |
| SOUND | ATK DEC REL VRAT | VDPT VDLY RVB CHOR |
| CC | CC16 CC17 CC18 CC19 | CC20 CC21 CC22 CC23 |

# 6 Filters

Every filter cell uses the same words: `TYPE` is the model, `MODE` its response, then `CUT` and `RESO`. While you turn any filter knob, the response curve is drawn over the half of the grid you are not using; `FLTR` shows it full-size.

**Envelopes have a picture too.** On any page with an envelope (`AMP`, `F ENV`, `MENV`, an FM operator's page), turning `ATK`, `DEC`, `SUS`, `REL` or `CURV` draws the envelope's shape over the other half of the grid for a second: the rise, the fall to the sustain level, a held stretch, and the release, with dotted lines where one stage ends and the next begins. `CURV` bends the lines exactly as it bends the sound. Each stage is drawn wider the longer it lasts, but on a compressed scale (a stage ten times longer is a few columns wider, not ten times), so a 1 ms attack and a 10 s release can share the picture.

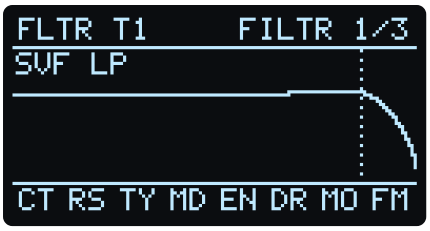
**Each stage can be bent on its own.** Hold `Shift` and turn `ATK`, `DEC` or `REL`: instead of a fine change of the time, that one stage changes shape. Turned left (down to −100) the move comes early: an attack that bulges upward, a decay or release that drops at once and then levels out. Turned right (up to +100) it comes late: a hollow attack, a decay or release that holds and then falls. 0 is a straight line. While you turn, the cell shows the curve (`ACRV`, `DCRV` or `RCRV`), the name bar says for example "ATK CRV +40", and the picture shows the new shape; a second later the cell shows the time again. This works wherever those three cells belong to an envelope: `MAIN`, `AMP`, `F ENV`, `MENV`, every FM operator, on `FLTR`, and on a held step, where it locks the curve for that step. `SUS` is a level and has no shape, so `Shift` stays fine there; a MIDI track's `ATK`, `DEC` and `REL` are controller values and have no curve either. Where there is a `CURV` cell it still bends all three stages together, and each stage's own bend is added to it. The curves are saved with the sound.

| TYPE | WHAT IT IS | MODE | CHARACTER |
|---|---|---|---|
| `SVF` | 12 dB state-variable | LP BP HP NTCH | Clean, stable right up to the top; the neutral choice. |
| `SEM` | Expander-style 12 dB | MRPH BP | Gentle and wide. In MRPH, `MORP` sweeps low-pass through notch to high-pass. Never self-oscillates. |
| `LADR` | Transistor ladder | LP24 LP18 LP12 LP6 BP HP | The classic thick low-pass. Self-oscillates at full resonance. |
| `DIOD` | Two-pole with diode limiting | LP HP BP | Rougher, asymmetric resonance. |
| `COMB` | Tuned feedback comb | POS NEG | Metallic, pitched resonances; cutoff sets the pitch. |
| `OFF` | Bypassed | – | – |

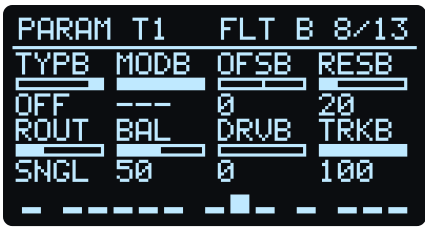
`DRIV` overdrives the filter input, each type in its own way (the SVF soft-clips; on COMB it darkens the feedback). `MORP` acts in one place, type SEM in mode MRPH; anywhere else the knob says `SEM MRPH ONLY`. `ENV` sets how far the filter envelope moves the cutoff, up to seven octaves either way. `KTRK` / `KEY` makes the cutoff follow the keyboard.

**The synth has two filters.** The `FLT B` page sets the second one: its own type and mode, its cutoff as an offset from filter A (`OFSB`), and `ROUT`: single, series or parallel, with `BAL` to balance them.

**FM, macro, wavetable and wave sequencer** share one filter design (`FILTR` and `F ENV` pages), open by default so presets sound as they did before you touch it. **The sampler** has its own `FILTR` page. A filter is also available as an insert effect (`FLT`) for any track, drums included.




FLTR




Synth, FLT B

# 7 Modulation and pressure

## LFOs

LFOs run per voice: each note has its own. The synth has up to four, on its `MOD` and `MOD2` pages (choose which with `LFO`, how many with `LFOS`). Every other sound engine ends with three pages: `STRIP` (what the mod strip does, see Strips below), then `LFO1` and `LFO2`.

| CELL | WHAT IT DOES |
|---|---|
| `RATE` | Speed in Hz. With `SYNC` on, a tempo division instead (8 bars to 1/32, dotted and triplet included); both values are remembered. |
| `DPTH` | Amount; negative inverts. |
| `SHPE` | `SIN` `TRI` `SQR` `SAW` `RND` `SRND` `RAMP` `EXPU` `EXPD` — sine, triangle, square, saw, stepped random, smooth random, ramp, and two exponential shapes that only push one way. |
| `TRIG` | `FREE` runs on regardless of notes. `KEY` restarts with each note. `ONE` plays one cycle and stops: a one-shot envelope in any shape. `HALF` plays half a cycle. `HOLD` takes one value at note-on and keeps it: a different value per note. |
| `DEST` | What it moves; see below. |
| `FADE` | Fade-in time after the note starts. |
| `PHAS` `DLY` | Synth only: start phase, and a wait before the LFO begins. |
| `WHL` `VEL` `PRES` | Synth only: the mod strip, velocity, and the note's own pad pressure add to the depth. `DPTH` 0 with `WHL` up gives vibrato from the strip alone. |
| `KTRK` | Synth only: higher notes speed the LFO up. |

### Destinations

An LFO can reach nearly everything its engine has a knob for. `DEST` starts with the engine's basic destinations (pitch, level, pan, cutoff, resonance and the engine's main controls) and goes on with every other continuous cell of the engine's pages, page by page; on the six engines with two LFOs the list ends with the other LFO's rate and depth (`LRAT`, `LDPT`). Plain turns go one entry at a time; `Func` + turn jumps to the start of the next page's group, so the far end of a long list is a few clicks away.

A destination that is a page cell has that cell's name, or the name with a number or an F in front where two pages use the same word (`ATK1` is operator 1's attack, `FATK` the filter envelope's). The LFO moves the cell around the value its knob is set to, in the cell's own scale: at `DPTH` 100 a 0–100 cell swings by its whole range and a time or frequency cell by half its range in octaves, never past the cell's limits. Turn the knob and the swing moves with it. Every note has its own LFOs, so in a chord each note's cell moves by itself.

| ENGINE | DEST OFFERS (AFTER OFF) |
|---|---|
| Synth | 74: `PTCH` `CUT` `RESO` `MIX` `AMP` `PW` `XMOD` `CUTB` `PAN` `LVL1` `LVL2` `LVL3` `NOIS` `FFM` `PIT2` `RAT1` `RAT2` `RAT3` `RAT4` `DPT1` `DPT2` `DPT3` `DPT4` · `TUN2` `DET2` `DRIV` `TUN3` `NCOL` · `1A` `1B` `2A` `2B` `3A` `3B` · `SPRD` `GLID` `DRFT` `DRAT` · `ENV` `FDRV` `MORP` `RESB` `BAL` `DRVB` `TRKB` · `FATK` `FDEC` `FSUS` `FREL` `KEY` `FVEL` `PMOD` · `ATK` `DEC` `SUS` `REL` `VEL` · `ATK2` `DEC2` `SUS2` `REL2` · `ATK3` `DEC3` `SUS3` `REL3` · `ATK4` `DEC4` `SUS4` `REL4` · `ATK5` `DEC5` `SUS5` `REL5` `EDPT` |

| ENGINE | DEST OFFERS (AFTER OFF) |
|---|---|
| FM | 68: `PTCH` `AMP` `PAN` `CUT` `RESO` `INDX` `FDBK` `DETN` `LVL1` `LVL2` `LVL3` `LVL4` `LVL5` `LVL6` `RAT1` `RAT2` `RAT3` `RAT4` `RAT5` `RAT6` `DET1` `FDB1` `ATK1` `DEC1` `SUS1` `REL1` `DET2` `FDB2` `ATK2` `DEC2` `SUS2` `REL2` `DET3` `FDB3` `ATK3` `DEC3` `SUS3` `REL3` `DET4` `FDB4` `ATK4` `DEC4` `SUS4` `REL4` `DET5` `FDB5` `ATK5` `DEC5` `SUS5` `REL5` `DET6` `FDB6` `ATK6` `DEC6` `SUS6` `REL6` `ENV` `DRIV` `MORP` `KTRK` `FATK` `FDEC` `FSUS` `FREL` `EVEL` `FVEL` `LRAT` `LDPT` |
| Macro | 26: `PTCH` `AMP` `PAN` `CUT` `RESO` `SAWS` `SPRD` `HARM` `MORP` `LEVL` `ATK` `DEC` `SUS` `REL` `ENV` `DRIV` `FMOR` `KTRK` `FATK` `FDEC` `FSUS` `FREL` `EVEL` `FVEL` `LRAT` `LDPT` |
| Wavetable | 33: `PTCH` `AMP` `PAN` `CUT` `RESO` `POS` `SPRD` `SUB` `MSPD` `MDPT` `ENVP` `VELP` `TIME` `SPD` `GLID` `VIBR` `ATK` `DEC` `SUS` `REL` `VEL` `ENV` `DRIV` `MORP` `KTRK` `FATK` `FDEC` `FSUS` `FREL` `EVEL` `FVEL` `LRAT` `LDPT` |
| Wave seq | 74: `PTCH` `AMP` `PAN` `CUT` `RESO` `RATE` `XFAD` `PRNG` `LEVL` `ATK` `DEC` `SUS` `REL` `VEL` `ENV` `DRIV` `MORP` `KTRK` `FATK` `FDEC` `FSUS` `FREL` `EVEL` `FVEL` `LRAT` `LDPT` `P1` `P2` `P3` `P4` `P5` `P6` `P7` `P8` `P9` `P10` `P11` `P12` `P13` `P14` `P15` `P16` `L1` `L2` `L3` `L4` `L5` `L6` `L7` `L8` `L9` `L10` `L11` `L12` `L13` `L14` `L15` `L16` `T1` `T2` `T3` `T4` `T5` `T6` `T7` `T8` `T9` `T10` `T11` `T12` `T13` `T14` `T15` `T16` |
| Sampler | 41: `PTCH` `AMP` `PAN` `CUT` `RESO` `STRT` `DRIV` `GPOS` `SIZE` `DENS` `SPD` `END` `LPST` `LPEN` `SCAN` `SPRY` `PJIT` `SPRD` `RVRS` `SLIC` `FADE` `SLEN` `SSIZ` `XFAD` `JIT` `SSPR` `BITS` `RATE` `LEVL` `VELF` `ENVF` `KTRK` `FDRV` `ATK` `DEC` `SUS` `REL` `VEL` `RAND` `LRAT` `LDPT` |
| Drums | 17: `PTCH` `DEC` `TONE` `DRIV` `SNAP` `AMP` `PAN` `SWEP` `SWPT` `SCRV` `SHAP` `KNCK` `KTON` `HOLD` `ACRV` `LRAT` `LDPT` |

On the synth, `1A` to `3B` are the two `SRC` cells of each oscillator slot; `ATK2` to `REL5` and `EDPT` are the extra envelopes of the `MENV` page. On the wave sequencer, `P1` – `P16`, `L1` – `L16` and `T1` – `T16` are the pitch, level and time of one step.

> **STRUCK VALUES** Some values are read once, when the note is struck, so the LFO sets them per note rather than moving them while the note sounds. Use `FREE` or `HOLD` there; `KEY` would give the same value every time. Drums: `DEC`, `SNAP`, and on a kick-machine pad `SWPT`, `SCRV`, `KNCK`, `HOLD`, `ACRV`. Sampler: `STRT`, `END`, `LPST`, `LPEN`, `SLIC`, `FADE`, `SLEN`, `VELF`, `KTRK`, `ATK`, `DEC`, `SUS`, `REL`, `RAND`; its grain cells `SPRY`, `PJIT`, `SPRD`, `RVRS` are read each time a grain starts.

> **NOT IN THE LISTS** Cells that choose from a list or count something (types, modes, voices, steps), cells that rebuild a sample or a table, whole-sound tuning (`PTCH` is the pitch destination), envelope curves, and `LEVL` where it is the level of the mixed voices (synth, FM, wavetable: use `AMP`).

## Extra envelopes

The synth's `MENV` page adds up to four envelopes beyond the amp and filter ones: `ENV` selects, `ENVS` sets how many run, then `ATK DEC` | `SUS REL DPTH DEST`, with destinations amp, cutoff or pitch and then the same long list as the LFOs.

## Pad pressure

Each pad sends its own pressure, and it reaches only the note that pad is playing. There are three ways to use it.

### 1. Routes, per track

`AT ROUTE` gives each track up to four routes: a destination and a depth. A new track starts with pressure on level (drums: decay).

| ENGINE | PRESSURE DESTINATIONS |
|---|---|
| Synth | LEVL CUT VIB PW MIX XMOD |
| FM | LEVL CUT FM |
| Macro | LEVL CUT VIB TIMB MORP |
| Wavetable | LEVL CUT VIBR POS |
| Wave seq | LEVL CUT |
| Drums | DEC TUNE TONE LEVL DRIV SNAP PAN (one route for the kit) |
| Sampler | none (use ranges, below) |
| MIDI track | sends per-note pressure out |

### 2. Ranges, on any knob

On `PARAM`, `TRK FX` or `FLTR`: turn a knob to a value, then `Func` + ▲ sets that as the high end of a pressure range, or `Func` + ▼ as the low end. The name bar confirms (`RESO HI 80`). The bottom-row pad under that knob turns violet; pressing into it sweeps the knob and lets it return on release. With one end set, pressure moves from the knob's value to that end; with both, from low to high. The same gesture again clears the end.

### 3. The arp and rolls

Pressure can drive the arp's rate, octave range or gate (`ACFG` page) and the speed of pad rolls (`ROLL`).

## Strips

The pitch strip bends the selected track by its engine's `BEND` range (a drum kit: two semitones, the whole kit together) and springs back. It acts on the selected track only: other tracks playing from the sequencer stay in tune.

The mod strip holds its position. It always does something to the selected track, on every engine and every factory preset, and nothing at all when it is at the bottom. What it does is yours to change. On FM, macro, wavetable, wave sequencer, sampler and drums the `STRIP` page (just before `LFO1`) has three routes, `DST1` `AMT1`, `DST2` `AMT2`, `DST3` `AMT3`: each sends the strip to one destination of the list the LFOs use (Func + turn jumps from group to group) by a signed amount in that destination's own scale, so +100 with the strip fully up is where an LFO at full depth would peak. Pitch goes by the amount squared, up to an octave. Routes to the same destination add up, `LRAT` and `LDPT` from the strip act on both LFOs, and an amount of 0 or a destination of `OFF` switches a route off. `WVIB` is vibrato the strip brings in (semitones at the top of the strip) and `VRAT` its speed. The amounts can be locked per step. Until you change the page, the strip does this:

| ENGINE | THE MOD STRIP BY DEFAULT |
|---|---|
| Synth | Goes to every LFO whose `WHL` is set (`MOD2` page), adding to that LFO's depth. While no LFO in use has a `WHL` amount it is vibrato, up to half a semitone. An LFO set to `EXPU`, `ONE`, 40 Hz, `DPTH` 0 passes the strip straight on to its `DEST`: that is how the factory presets and the three `ARCH` sounds open a filter or raise cross-mod from the strip. |
| FM | `DST1` `INDX` +100: raises the modulation index, up to three times (brighter, harder; from the strip `INDX` counts double). `WVIB` 0.25: a little vibrato. It also moves the cutoff by `WHL` on the `F ENV` page, as on the next three engines. |

| ENGINE | THE MOD STRIP BY DEFAULT |
|---|---|
| Macro | `DST1` the fourth macro ( `MORP` in most models) +100: sweeps it to its top. `DST2` the first macro +30. `WVIB` 0.25. Cutoff by `WHL` . |
| Wavetable | `DST1` `POS` +60: scans the table forward from `POS` , up to 60 % of the table. `WVIB` 0.25. Cutoff by `WHL` . |
| Wave sequencer | `DST1` `RATE` +50: runs the sequence faster, twice as fast at the top (so the top of the strip is in time again). Cutoff by `WHL` . |
| Sampler | Wears the sound down: `DST1` `DRIV` +70, `DST2` `RATE` +30 (sample-rate reduction, `TONE` page), `DST3` `SPRY` +50 (scatters the grains in `GRAN` mode). Whatever the strip adds to `DRIV` is mostly taken out of the level again, so the tone changes more than the loudness. |
| Drums | Opens the whole kit up: `DST1` `DEC` +79, every drum rings up to three times as long (from its next hit), and `DST2` `DRIV` +50, driven harder. The routes belong to the kit: every pad follows them. |
| MIDI | Sends controller 1 to the instrument outside. |

The strip moves the selected track only, and every track keeps the value the strip last gave it. Set one track's strip, select another and move the strip there: the first stays as you left it. Selecting a track changes nothing until you touch the strip, and a preset you load on a track is given that track's value again. While a track's arp is on, the strip can drive one arp macro instead ( `STRP` on the arp's `VARY` page); the track's sound then keeps whatever the strip last gave it.

# 8 Effects

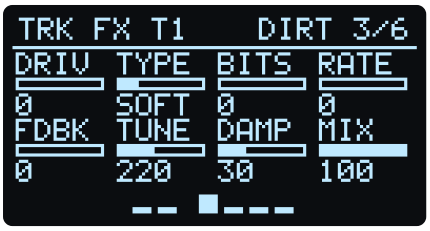
## Signal path

Engine → `DIRT` → `TALK` → `MODFX` → insert slots → mixer (volume, pan) → master EQ, compressor, limiter. Each track also sends to two shared buses. All of a track's effect pages are on `TRK FX`.

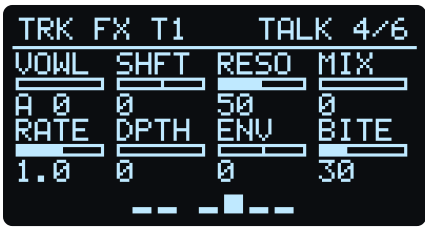
## The three character blocks

On every track, always available, bypassed while at zero.

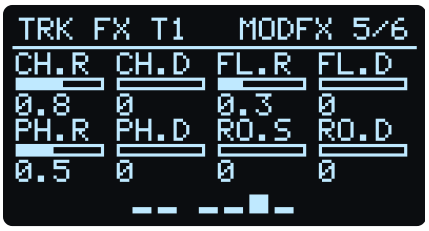
- `DIRT`: `DRIV TYPE BITS RATE` | `FDBK TUNE DAMP MIX`. Distortion with four curves, bit and sample-rate reduction, and a tuned feedback path that turns it into a resonator.
- `TALK`: `VOWL SHFT RESO MIX` | `RATE DPTH ENV BITE`. A vowel filter. The vowel can be moved by hand, by its own LFO, or by how hard you play.
- `MODFX`: chorus, flanger, phaser and rotary speaker, a rate and a depth each (`CH.R CH.D` | `FL.R FL.D` | `PH.R PH.D` | `RO.S RO.D`).



DIRT



TALK



MODFX

## Insert effects

The `FX` page shows one insert slot: `SLOT` chooses which, `TYPE` the effect, `WET` how much of it, and knobs 5 to 8 are the effect's own four controls.

> **CAREFUL** Turning `TYPE` replaces the effect and its four values at once, without the four-click confirmation. Undo (`Func` + `Shift` + `Play`) brings the effect back as it was; a scroll through the list counts as one step.

| TYPE | EFFECT | KNOBS 5-8 |
|---|---|---|
| `CHR` | Chorus | `RATE` `DPTH` `MIX` `VOIC` |
| `DRV` | Drive | `DRIV` `TONE` `MIX` `ALGO` |
| `FLG` | Flanger | `RATE` `DPTH` `FDBK` `MIX` |
| `PHS` | Phaser | `RATE` `DPTH` `FDBK` `STGS` |
| `TRM` | Tremolo | `RATE` `DPTH` `SHPE` `STER` |
| `DLY` | Delay | `TIME` `FDBK` `MIX` `MODE` |
| `RVB` | Reverb | `DCAY` `DAMP` `SIZE` `MIX` |
| `EQ` | Three-band EQ | `LOW` `MID` `MFRQ` `HIGH` |
| `CMP` | Compressor | `THR` `RATI` `ATK` `MAKE` |

| TYPE | EFFECT | KNOBS 5–8 |
|---|---|---|
| LMT | Limiter | CEIL REL LOOK |
| FLT | Filter | CUT RESO TYPE MODE |
| ENS | Ensemble | RATE DPTH WDTH MIX |
| PAN | Auto-pan | RATE DPTH SHPE PHAS |
| ROT | Rotary speaker | SPD HORN DRUM ACCL |
| WSH | Waveshaper | DRIV CURV TONE ASYM |
| RED | Rate reducer | RATE BITS FILT MIX |
| HALL | Hall reverb | DCAY SIZE DAMP PDLY |
| SPRG | Spring reverb | DCAY DISP TENS DAMP |
| SHIM | Shimmer reverb | DCAY SHIM MODE DAMP |
| RVSE | Reverse delay | TIME FDBK HCUT MIX |
| TRNS | Transient shaper | ATK SUS SPD OUT |
| RING | Ring modulator | FREQ RATE DPTH MIX |
| VIB | Vibrato | RATE DPTH SPRD FADE |
| WAH | Wah | FREQ RESO SENS RATE |
| AMP | Amp simulator | GAIN TONE CAB LEVL |
| GEQ | Four-band graphic EQ | LOW LMID HMID HIGH |
| TAPE | Tape delay | TIME FDBK AGE MIX |
| LCR | Left-centre-right delay | TIME FDBK SPRD MIX |
| MTAP | Multi-tap delay | TIME TAPS FDBK MIX |
| DUCK | Sidechain compressor | SRC THRS RATI RELS |

## Tempo sync

An insert that has a TIME (the delays) or a RATE (chorus, flanger, phaser, tremolo, auto-pan, ensemble, ring modulator, vibrato, wah) shows SYNC in cell 3 in place of ON. With SYNC on, that cell picks one of 22 divisions from 1/128 to 16 bars, triplets and dotted included, and follows the tempo. The free value is kept for when you switch sync off. A division too long for the effect is halved until it fits. To take a synced effect out, turn WET to 0.

## Sends and master

SND1 and SND2 on each track feed two buses shared by all tracks: a reverb and a delay to begin with. BUS1 and BUS2 on the TRACK page change what each bus carries, from the same list of effect types.

The master EQ, compressor and limiter are on SETTINGS → MASTR; master volume is MVOL on SETUP.

### Sidechain ducking

The DUCK insert turns its track down when another track plays: SRC chooses which (SELF makes it an ordinary compressor), THRS and RATI how hard, RELS how fast it comes back.

# 9 The sequencer

## Patterns and tracks

A project has 16 patterns. Each pattern holds the steps of every track; each track has its own length ( `LN` , 1 to 1024 steps) and speed ( `SP` : quarter, half, normal, double, quadruple). A track has one sound for all patterns.

Hold `>` and press a pad to queue a pattern. While playing, the change happens at the end of the loop; stopped, at once.

## Steps

On `STEP` , a tap adds or removes a step and a held step can be edited. Hold a step and:

- press keys to set its note or chord;
- keep holding: the display changes to the track's `MAIN` page showing that step's values, and any knob you turn is locked on that step;
- use the knob-page buttons to reach the other sound pages, and, one press back from `MAIN` , its `TRIG` page: `NOTE` , `VEL` , `LEN` (length), `MICR` (micro-timing, up to half a step early or late), and the condition cells.

## Parameter locks

A lock is a knob value that applies to one step only. Hold a step on `STEP` , `PARAM` , `TRK FX` or `FLTR` and turn a knob. While the step is held the cells show that step's values and the name bar shows its number ( `S5` ). After the step, the knob returns to its normal value. Hold several steps and the same turn locks all of them to the value shown. Engine controls, mixer levels and sends, effect blocks and inserts, and the arp's gate, velocity and ratchet can all be locked.

Remove one lock: hold the step (or several), hold `>` , turn the knob. Remove all of a step's locks: `LCKS` on its `TRIG` page, anticlockwise (an undo step).

**Pressure into locks.** On a step with recorded pad pressure, `>LCK` on its `PRES` page (four clicks) writes the swell as locks: on every step the note spans, each cell the track's pressure routes point at is locked to where the average pressure over that step would hold it. A route's depth is how far full pressure moves the cell from its own value toward its top (toward its bottom when the depth is negative). Steps under the note that had no trig become trigs that carry only locks, with no note of their own. Chords on a synth track share the cells, so their pressures are averaged; on a drum track each pad's own cell is locked. A route with no cell to lock (vibrato; `FM` on the FM synth; `TIMB` on the macro oscillator) is left out and the name bar says so ( `2 OF 3 ROUTES` ); if no route can be locked, nothing is changed ( `NO ROUTE` ). The recorded pressure is then removed: from here on the swell is edited like any other lock.

## Trig conditions

| COND | THE STEP PLAYS... |
|---|---|
| `ALL` | every time |
| `PROB` | with the chance set by `PROB` |
| `A:B` | on pass `A` of every `B` passes of the track (1:2 = every other time, 4:4 = every fourth) |
| `FILL` / `!FIL` | only while fill is held (`Shift` + `Rec` ) / only while it is not |
| `1ST` / `!1ST` | only on the first pass after starting / on every pass but the first |

| COND | THE STEP PLAYS... |
|---|---|
| `PRE` / `!PRE` | only if the previous conditional step played / only if it did not |

## Recording

**Start.** Hold `Rec` and press `Play`: recording goes on and the sequencer starts from the top. If it is already playing it keeps playing and recording simply switches on. `Rec` alone switches recording on or off without touching the transport (it acts when you let the button go), so `Rec`, then `Play` works too.

**Metronome.** While recording runs a click marks every beat, higher and louder on the first beat of each bar. `CLCK` on `SETTINGS` → `CLICK` chooses when it sounds: `OFF`, `REC` (only while recording runs: the setting it starts with) or `PLAY` (whenever the sequencer plays). `CLVL` sets its level; each turn of that knob plays one click, so you can set it before you start. The click follows the tempo, stops with the sequencer, and is only for your ears: it does not pass through the master effects or `MVOL`, and it is never recorded when you resample a track or the master.

**Count-in.** Set `CNT` on the same page to `1BAR` or `2BAR` and a recording starts with that many bars of clicks: `Rec` + `Play` (or `Play` with recording already on) shows `COUNT IN 1 BAR`, the metronome counts four beats to the bar – also when `CLCK` is off – and the sequencer and the recording start together on the beat after. During the count-in nothing plays, nothing you play is recorded, and a queued pattern waits. `Play` during the count-in calls it off. The count-in is only for a recording that starts the sequencer: if the sequencer is already playing, recording switches on at once, and `Play` without recording always starts at once.

**Knob movements are recorded too.** On `PARAM`, `TRK FX` and `FLTR`, a knob turned while the pattern records and plays is written into the selected track as parameter locks: one on every step that passes, from the first turn for one lap of the track, so a filter sweep comes back exactly as you made it. A knob you stop turning keeps writing its value until the lap is full. A step that had no trig becomes a trig without notes that only carries the lock. Afterwards the track's own value is what it was before you touched the knob; hold a step and `>` + turn removes a lock, and undoing the recording pass removes the whole movement. Sample choice, slicing and wavetable key frames cannot be locked and are not recorded.

**While recording** the name bar shows `REC FREE` or `REC QUANT` on every screen (a single `R` or `Q` when room is short) and a red light runs across the pads, one pad per step of the selected track. Pads that play notes keep playing them under the light; on `STEP` the playhead is the light, and recorded notes appear on the pads as you play them. Keys are written into the selected track, pad hits (each repeat of a roll included) into the track the pads play, arp output into the arp's track. Notes played together land on one step. A later pass adds to what is there; playing on a step that already holds notes replaces them.

**Pressure is recorded with pad notes.** What you do with a pad after striking it (press harder, ease off) is recorded with that note, for as long as you hold the pad, and the note is given the length you held it. On playback each note gets its own pressure back, through the track's pressure routes ( `AT ROUTE` ) as they are set at that moment, or as polyphonic pressure on a MIDI track: two pads held together keep their own swells. The pressure follows the tempo, the track's speed, micro-timing and swing with its note, and is let go when the note ends, when the track is muted and when the sequencer stops. A pad played by hand over a recorded note of the same pitch has the say while it is held. A note you do not press into stores nothing. A pad that is rolling ( `ROLL` ) uses its pressure for the rate of the roll, so that pressure is not recorded; with a roll threshold set, a pad held below the threshold is an ordinary note and its pressure is recorded. Keys have no pressure. A note can be at most one pass of its track long; a pad held longer is recorded up to there. Recording over a step replaces its notes and their pressure. See `PRES` on a held step's pages to scale, convert or remove what was recorded.

**Free or quantized.** Press `Rec` + `Play` again while recording to change between the two; the choice is kept for the next recording. `REC FREE` writes each note on the nearest step and keeps its distance from the grid as micro-timing,

so it plays back as you played it; QN on STEP pulls playback toward the grid by any amount and can be turned back down later. REC QUANT writes each note exactly on the nearest step.

**Stop.** Play stops the sequencer and ends recording; Play again plays back what you recorded. Rec alone ends recording and leaves the sequencer playing. Either way, if you recorded anything, STEP opens with the track you recorded last selected, on the step page that holds its first recorded step: hold a step and the display shows that track's MAIN page for it; turn a knob to lock a parameter on that step. (If pads are pinned, the pads stay the pinned ones until you unpin them.)

## Copy, keep, fill, tempo

- **Copy / clear / paste**: Func + Rec / Play / >, acting on the held step, the track ( STEP ), the pattern ( TRACKS , clear and paste held for half a second), the sound ( PARAM , FLTR ) or the effects ( TRK FX ).
- **Undo / redo**: Func + Shift + Play / Rec, fifteen steps back. A recording is one step.
- **Keep and reload**: > + Rec keeps a temporary copy of the pattern; > + Play brings it back, at once (it is meant to be played in time) and as an undo step. The copy lives until you quit.
- **Fill**: hold Shift + Rec; steps with the FILL condition play while you hold.
- **Tempo**: BPM on SETTINGS , knob 8 on STEP , or tap Func + Settings.
- **Swing**: SWNG on SETTINGS → SETUP .
- **Mute and solo**: bottom pads on TRACKS , or hold ▲ / ▼ and press a bottom pad from anywhere. A muted track keeps its place and comes back in time.

## Song mode

SONG is an ordered list of rows, each a pattern, a repeat count and a transposition for pitched tracks. Switch SG on and Play follows the list; LP loops it.

## Clock

CLK on SETTINGS sets MIDI clock: INT (Neptune is the clock), EXT (follow incoming clock) or AUTO . LINK is listed for the future and does nothing yet.

# 10 Arpeggiator and rolls

## The arpeggiator

Every track has its own. `Arp` opens the screen; pressed again there, it switches the selected track's arp on. `Shift` + `Arp` latches it so it carries on after you let go (with the arp off, the same gesture latches the keys themselves). Arps follow the transport, the scale lock and the tempo.

| PAGE | KNOBS 1–4 | KNOBS 5–8 | FOR |
|---|---|---|---|
| `ARP` | MODE RATE OCT GATE | RPT SWNG GLID LTCH | The basics |
| `PLAY` | STRM TIME DIR VEL | RTCH RDEC HUMN FREZ | Strumming, ratchets, humanize, freeze |
| `VARY` | SPCE DICE DENS SCAT | RCHN GVAR STRP VAR | Variation macros |
| `MUTA` | MUTA INTV MODE SPCE | EVOL ANC >PAT | Mutation over time |
| `ACFG` | DEST DPTH REC LOCK | TAP RHYT | Pressure, recording, tapped rhythm |
| `ROLL` | ROLL MODE BASE MAX | DPTH GATE P>V DCAY | Pad rolls |
| `ROL2` | SWNG GLID THRS LTCH | HUMN ACNT PROB RPTS | Pad rolls, continued |

Modes: Up; Down; Up-down; Random; As played; Chord repeat; Walk; Play order; Converge; Diverge; Poly arp (all notes, accented); Random with octave jumps; Parallel (PARA: every key runs its own climb through the octaves from the moment it is pressed, side by side with the others; keys pressed together climb together); Pedal (PEDL: the lowest note sounds before each higher one in turn – 1 2 1 3 1 4); Rising pedal (PDL+: the same, and after each pass the next note up becomes the pedal – 1 2 1 3 1 4, 2 3 2 4, 3 4).

### Variation

The `VARY` macros change what the arp plays without changing what you hold: `DENS` thins it out, `SCAT` throws notes up and down octaves, `RCHN` adds ratchets, `GVAR` varies note lengths, `SPCE` sets how bold all of that is and `DICE` re-rolls it. `STRP` hands one of them to the mod strip.

### Mutation

With `MUTA` on, the pattern changes itself every `INTV` bars. `MODE` decides whether each change starts from the ancestor again ( `ANC` ) or builds on the last ( `CUM` ). `EVOL` shows the generation: turn it up to mutate now, down to go back to the ancestor. `>PAT` (four clicks) writes what the arp is playing into the track's pattern as ordinary steps. Going back to the ancestor and `>PAT` can both be undone.

### Tapped rhythm

Switch `TAP` on and tap a rhythm on any pad (they light amber). Switch it off, switch `RHYT` on, and the arp plays its notes in your rhythm.

### Locking the arp

`TRK FX` → `ARPLK` holds the arp's gate, velocity and ratchet as lockable cells, so steps of the pattern can change how the arp behaves as it passes.

## Pad rolls

With `ROLL` on, any pad that plays a note ( `DRUMS` , `SCALE` , `AT ROUTE` , `PRESET` ) repeats while held. Pressure raises the rate from `BASE` to `MAX` : tempo divisions in `QUANT` mode, free Hz in `FREE` mode. `P>V` lets pressure set the loudness of each repeat; `DCAY` fades them.

`ROL2` refines it: swing, a glide between rates, a pressure threshold so light touches play single hits ( `THRS` ), latch, humanize, an accent every Nth hit, a probability per repeat and a limit on the number of repeats. Rolls are recorded hit by hit.

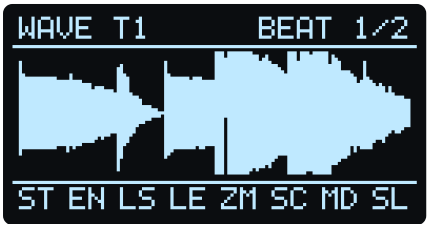
# 11 Sampling, slicing and stretch

## Samples

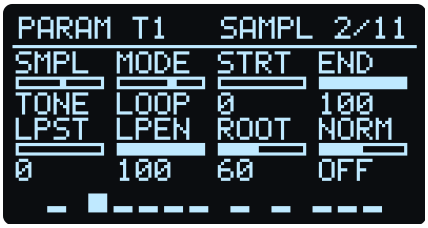
The sampler's `SMPL` cell chooses from the bank: a built-in drum loop, eight hits and a tone, any WAV or AIFF file in *Music / Neptune / Samples* (read at start-up), eight drum breaks (below), and your resampled takes. Set start, end and loop on the `SAMPL` page or by eye on `WAVE`; `Func` + turn is coarse and `Shift` + turn fine, and zooming ( `ZM` ) makes the markers finer still.

Play modes: `FWD` , `REV` , `1SHT` (plays to the end regardless of the key), `GATE` (stops on release), `LOOP` , `PING` (loops back and forth), `GRAN` .

`ROOT` is the key at which the sample plays at its recorded pitch; `NORM` brings a quiet sample up to full level.



WAVE



Sampler, SAMPL

## Drum breaks

Eight drum breaks come with the app, after your own files in the `SMPL` list. The name is the tempo the break was played at: B080, B090, B105, B120, B157 and B160 are recordings of drummers, released by them into the public domain (CC0; names and sources are in *docs/sample-credits.md*), and B100 and B170 are made by the app's own drum voices. Each is one, two or four whole bars. Choosing a break sets `LEN` on the `STRCH` page to its length, so turning `SYNC` on is all it takes to play it in time with the project.

Every break has two sampler presets on the `PRESET` screen (category PERC):

- **Break 90 Loop**: hold a key and the break goes round in time with the tempo, whatever tempo it was played at. It is cut at its drum hits ( `MODE` = `TRNS` ) and played with `ALGO` = `SLCE` , so each hit sounds as recorded and only its place in time moves; other keys play it higher or lower without changing the timing. One voice: a new key starts the break again.
- **Break 90 Slices**: the sixteen strongest hits, one on each key going up from C4 ( `PLAY` = `KEYS` , play mode `1SHT` ), at their recorded speed and pitch. The pads of `PRESET` play sixteen keys in a row from the C of the track's octave: at C4 that is one slice per pad. Record them into steps to build your own beat from the break.

From there everything on the sampler's pages applies: `ALGO` = `TAPE` for the repitched sound, `PLAY` = `RAND` or `NEXT` on the slices, `SLCS` for fewer or more of them.

## Slicing

The `SLICE` page cuts the sample into pieces you can play separately.

- `MODE` = `GRID` : `SLCS` equal slices. Right for a loop that is already in time.
- `MODE` = `TRNS` : one slice per hit, found by listening for the sudden rise at the start of each sound. `SENS` sets how quiet a hit may be and still count; `GAP` sets the shortest time between two hits. `SLCS` shows how many were

found, and turning it down keeps only the strongest. Every slice starts on a zero crossing, so slices do not click. Hits closer together than about a fiftieth of a second are heard as one.

`PLAY` decides which slice a note plays:

- `ONE` : the slice chosen with `SLIC` (or pad N on `WAVE` ). Lock `SLIC` per step to re-order a loop.
- `KEYS` : each key plays its own slice, going up from `ROOT` . The loop is laid out across the keyboard.
- `NEXT` : every new note plays the next slice, so a row of identical steps walks through the loop.
- `RAND` : a random slice per note.

`FADE` softens the end of each slice and `SLEN` shortens them all for a gated feel. The `WAVE` screen draws a tick at each slice and lights the pad of the slice that is sounding.

> **NOT LOCKABLE** `SMPL` , `NORM` and the cells that re-cut the sample ( `MODE` , `SLCS` , `SENS` , `GAP` ) cannot be locked to a step or driven by pressure: each change re-analyses the whole sample. `SLIC` , `PLAY` , `FADE` and `SLEN` can.




Sampler, SLICE

## Time stretch

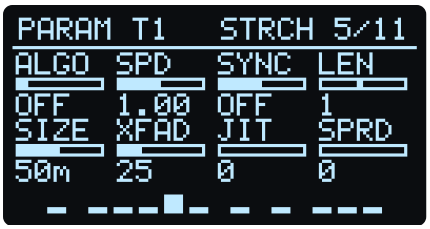
The `STRCH` page changes how long the sample lasts without (except in `TAPE` ) changing its pitch. The key still sets the pitch; start, end and loop points, play modes, filter, envelope and LFOs all work as usual. Each `ALGO` has a sound of its own, and that is the point of having several.

| ALGO | HOW IT WORKS | SOUNDS LIKE |
|---|---|---|
| `TAPE` | Plays faster or slower, like tape or vinyl. | Pitch drops as it slows. With SYNC, the classic repitched loop. |
| `CYCL` | Repeats short cycles of the sound, crossfading from one to the next. | The metallic, buzzing stretch of early hardware samplers. SIZE tunes the buzz. |
| `GRAN` | Overlapping grains from the granular engine. | Smooth and slightly diffuse. Good on pads, vocals, textures. |
| `SLCE` | Does not stretch anything: each slice plays untouched and is started when its turn comes. | Tight and punchy on drums. Slowing down leaves gaps between hits; speeding up cuts them short. |
| `SMER` | Very long grains from random nearby positions, at a sixteenth of the shown speed. | A frozen wash. A second of sound becomes a slowly moving chord. |
| `WSOL` | Like CYCL, but each cycle is lined up with the last by matching their waveforms. | The cleanest of the set on single notes and voices. |

`SPD` sets the speed from a quarter to four times. With `SYNC` on, the sample between its start and end markers is made to last exactly `LEN` (a sixteenth of a bar to 32 bars) at the current tempo, and follows the tempo if you change it: a loop recorded at any speed locks to the sequencer. `SIZE` , `XFAD` and `JIT` shape the grain of the stretch, and `SPRD` widens `GRAN` and `SMER` .

The LFO destination `SPD` moves the speed by up to two octaves either way: tape-stop dives, or a stretch that breathes.

In `GRAN` play mode the grain page is in charge and `ALGO` is ignored, but `SYNC` still works: `SCAN` at 1.00 then means the region lasts `LEN`.



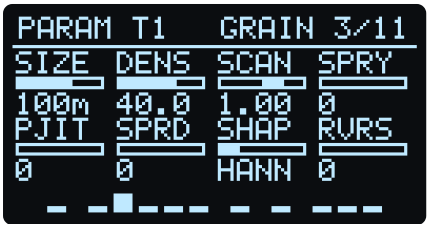

Sampler, STRCH

## Granular

In `GRAN` mode the sampler plays the sample as a cloud of short grains. Pitch comes from the key; time is separate. The `GRAIN` page:

- `SIZE` and `DENS`: how long each grain is and how many start per second. Long and dense is smooth; short and sparse is granular in the gritty sense.
- `SCAN`: how fast the read point travels through the sample. 1× is normal speed, 0 freezes on the spot, negative runs backwards. With `SCAN` at 0, `STRT` is the position: turn it, lock it, or put an LFO on `GPOS`.
- `SPRY`: scatters grain positions and timing. `PJIT`: random pitch per grain.
- `SPRD` stereo spread, `SHAP` the grain's envelope, `RVRS` the share of grains played backwards.

The `WAVE` screen shows each grain as a tick while it sounds.



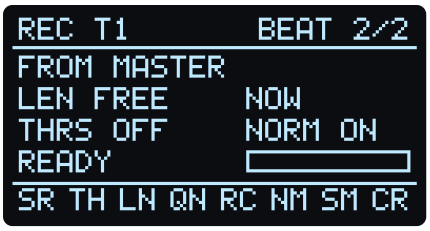

Sampler, GRAIN

## Resampling

The `REC` page of `WAVE` records Neptune's own sound into a new sample.

1. `SR`: the source, the master output or one track after its effects.
2. `LN`: free length, or a set length from one beat to eight bars. `QN`: start now or on the next bar. `TH`: optionally wait for the source to pass a level.
3. `RC`: two clicks clockwise within a second arms and starts; one click anticlockwise stops. The top pads are red while recording.
4. The take is saved as *R001.wav*, *R002.wav*… in *Music / Neptune / Samples* and given to the selected track: as its sample on a sampler track, as its `USER` table on a wavetable track (or, while that track's table is `KEYS`, as the sample of the key chosen on the `WAVE` page).

`NM` normalises takes, `SM` picks a different sample for the track, and `CR` (four clicks) crops the sample to its markers.




WAVE, REC page

# 12 Projects, presets and the rest

## Projects

A project is everything: tempo, all 16 patterns with their steps and locks, every track's engine and sound, effects, sends, master section, song, scale pages, pressure routes and ranges, and arp settings. There are 32 slots (*Music / Neptune / Projects*, P01 to P32).

`SETTINGS` → `PROJ` : `SLOT` chooses, `LOAD`, `SAVE`, `SVAS` (save as, to the chosen slot) and `NEW` each need four clicks and answer OK or FAIL. `AUTO` keeps saving to the current slot as you work. The app reopens the last project when it starts, and on top of it everything exactly as it was when the app was last running, saved or not (the app notes its state when it quits and every 15 seconds). `SAVE` is still what writes a project slot; `NEW` or `LOAD` is the way to a different starting point.

Not saved: an arp's mutation state, the temporary pattern copy, the clipboards, and the keyboard split.

## Presets

A preset is one track's sound. The built-in ones are marked * in the list, among them a loop and a sliced preset for each of the sampler's drum breaks (chapter "Sampling, slicing and stretch"); yours are saved from `PRESET` with `SV` into *Music / Neptune / Presets*. Categories: BASS, PAD, LEAD, KEYS, FX, PERC, KIT, INIT.

**What `SV` saves:** the track's own sound, complete - every engine setting, the track's insert effects and their settings, the DIRT / TALK / MODFX blocks, the send levels, the volume, and the track's arpeggiator (all its settings, whether it is on, and its tapped rhythm). If a preset is being auditioned at that moment, it is still your own sound that is saved, not the one you are listening to. A preset saved this way brings its effects and its arpeggiator with it when it is auditioned or loaded; the built-in presets have neither of their own and play through whatever effects and arp the track has.

**Naming:** `SV` saves at once as User 1, User 2 and so on, so nothing waits for the Mac. The Mac then shows a box asking for a name: type up to 15 characters on its keyboard and press Return, or press Escape to keep the User name. A name already in use is refused and the name bar says so.

## Keyboard split

`Shift` + `Oct−` drops the lowest twelve keys by one further octave, then two, then three, then back to normal; `Shift` + `Oct+` steps the other way. The name bar shows `SPLIT −1 OCT`. Plain `Oct−` / `Oct+` still shift the whole keyboard.

## Latency and headroom

Neptune adds about 1.3 ms of its own between a key and its sound. What you hear beyond that is the audio output: an interface or wired headphones add a few milliseconds, the Mac's built-in speakers about 19 ms.

The heaviest test session (eight tracks, eight notes each, every effect block plus delay and reverb on every track) uses about a quarter of one processor core on an M3 Pro. You are unlikely to run out.

The display redraws about twelve times a second, which is as fast as the Launchkey accepts pictures.

## If something seems wrong

| WHAT YOU SEE | WHAT TO CHECK |
|---|---|
| No sound | `MVOL` on `SETTINGS`; the track's `VOL`; whether it is muted (red pad on `TRACKS`) or another track is soloed (blue). |
| Pads do not play the drums | Only `DRUMS` has drum pads: `Shift` + bottom pad 2. The keys play the selected drum track on every screen. |
| A knob does not seem to change the sound | A step lock or a pressure range may be overriding it. Orange pads on `PARAM` have locks; a violet pad has a range. |
| Pads play notes on a screen where they should not | They are pinned (`PIN` in the name bar). On `DRUMS` or `SCALE`, hold `Shift` and hold that screen's bottom pad for half a second. |
| Notes hang or the arp will not stop | Latch is on (`LATCH` in the name bar): `Shift` + `Arp`. |
| A sound or an effect rings on and will not stop | `Shift` + `Play` stops the sequencer, lets every note go and cuts every effect's tail. |
| Low keys are in the wrong octave | A keyboard split is on: `Shift` + `Oct+` until the `SPLIT` message says off. |
| Copy or paste did nothing | They act only on `STEP`, `TRACKS`, `PARAM`, `FLTR` and `TRK FX`, or on a held step. On `TRACKS` clear and paste need the button held for half a second. |
| A synced delay is shorter than the division says | The division was longer than the delay can hold and has been halved to fit. |
| The display is blank after plugging in | Quit and restart the app with the Launchkey already connected. |

# Appendix A Short names

Every short name the display can show on a knob, in alphabetical order. **Belongs to** names the engine, effect or screen and, after the colon, the page. Names that mean different things in different places have one row per meaning. Values shown as 0–100 are the display's scale.

| SHORT | FULL NAME | BELONGS TO | VALUES | WHAT IT DOES |
|---|---|---|---|---|
| A | A of A:B | STEP: TRIG | 1–8 | With COND = A:B the step plays on pass A of every B passes. |
| ACCL | Acceleration | Insert effect: ROT | 0–100 | How quickly the rotors change speed. |
| ACNT | Accent | ARP: ROL2 | every Nth | Accents every Nth repeat. |
| ACRV | Kick decay curve | Drums: KICK | −100…+100 | Kick machine: shape of the decay; negative falls fast then lingers. |
| AGE | Tape age | Insert effect: TAPE | 0–100 | Wow, flutter and dulling of the tape. |
| AL | Algorithm | WAVE: WAVE (wavetable track) | LIN … VOC | How the frames between two keys are made (ALGO). |
| ALGN | Align | Wavetable: KEYS | OFF ON | On: every key is turned in time until its fundamental starts like a sine, so waves cut from different places do not cancel each other. Off: as they were cut. |
| ALGO | Algorithm | FM: FMMIX | 6>1 2+3 3x2 2x3 1>3 ADD 2+4 4+2 | How the six operators are wired (8 algorithms, from one chain of six to six parallel carriers). |
| ALGO | Drive algorithm | Insert effect: DRV | SOFT HARD FOLD CRSH | SOFT, HARD, FOLD or CRSH (bit-crush). |
| ALGO | Stretch algorithm | Sampler: STRCH | OFF TAPE CYCL GRAN SLCE SMER WSOL | OFF, TAPE (pitch follows speed), CYCL (vintage cyclic stretch, metallic), GRAN (smooth granular), SLCE (slices keep their sound, only their timing moves), SMER (extreme smear), WSOL (clean stretch). |
| ALGO | Key-frame algorithm | Wavetable: KEYS | LIN STEP CLSC PHAS TRNS VOC | How the frames between two keys of the KEYS table are made. LIN: a crossfade. STEP: none, the nearer key. CLSC: a crossfade kept as 128 samples of 8 bits. PHAS: level and phase of each harmonic separately, so nothing cancels on the way. TRNS: the energy glides along the harmonics (optimal transport), so a formant slides instead of fading. VOC: first the outline of the spectrum changes, then its harmonics; half way it is one key's formants on the other's harmonics. |
| AMP | Level | all engines: LFO DEST | destination | The voice's level, from silence to double: tremolo. |
| AMT1 | Strip route 1 amount | FM, Macro, Wavetable, Wave seq, Sampler, Drums: STRIP | −100…+100 | STRIP page: how far the strip, fully up, moves the destination of route 1, in that destination's own scale: +100 is what an LFO at full depth reaches at its peak. Negative moves it the other way; 0 switches the route off. |
| AMT2 | Strip route 2 amount | FM, Macro, Wavetable, Wave seq, Sampler, Drums: STRIP | −100…+100 | STRIP page: how far the strip, fully up, moves the destination of route 2, in that destination's own scale: +100 is what an LFO at full depth reaches at its peak. Negative moves it the other way; 0 switches the route off. |

| SHORT | FULL NAME | BELONGS TO | VALUES | WHAT IT DOES |
|---|---|---|---|---|
| AMT3 | Strip route 3 amount | FM, Macro, Wavetable, Wave seq, Sampler, Drums: STRIP | −100…+100 | STRIP page: how far the strip, fully up, moves the destination of route 3, in that destination's own scale: +100 is what an LFO at full depth reaches at its peak. Negative moves it the other way; 0 switches the route off. |
| ANC | Ancestor | ARP: MUTA | ORIG / KEPT | What mutations return to: the original pattern or the last kept one. |
| ARCH | Architecture | Synth: VOICE | INIT MONO HYBR POLY | Loads a starting voice architecture into the same pages: INIT, MONO, HYBR (hybrid) or POLY. Each one keeps what you dial in on it: switch away and back and the sound is as you left it. |
| ASYM | Asymmetry | Insert effect: WSH | 0–100 | Bends the curve to one side for even harmonics. |
| ATK | Attack | Insert effect: CMP | 0.1–100 ms | How fast the compressor reacts, in milliseconds. |
| ATK | Attack gain | Insert effect: TRNS | −100…+100 | Boosts or softens the start of each hit. |
| ATK | Attack (CC 73) | MIDI track: SOUND | 0–127 | Sends the attack-time controller. |
| ATK | Attack | Synth: F ENV, AMP; FM: OP1–OP6, F ENV; Macro, Wavetable, Wave seq: AMP, F ENV; Sampler: AMP | varies | Time the envelope takes to rise to full level. |
| ATKn | Envelope attack | Synth, FM: LFO DEST | destination | FM, ATK1 to ATK6: the attack of operator 1 to 6. Synth, ATK2 to ATK5: the attack of an extra envelope (MENV). |
| AU | Audition | PRESET: bar | OFF / ON | On (the default): the highlighted preset takes over the track's sound for listening only; the previous sound returns on leaving unless LD is used. Off: the list scrolls without changing the sound. |
| AUTO | Autosave | SETTINGS: PROJ | OFF / ON | Keeps saving the full state to the current slot. |
| B | B of A:B | STEP: TRIG | 1–8 | The cycle length for A:B. |
| BAL | Filter balance | Synth: FLT B | 0–100 | Balance between filter A and filter B. |
| BASE | Base rate | ARP: ROLL | division or Hz | Repeat rate at the lightest pressure. |
| BEND | Bend range | Synth: OSC3; FM: RATIO; Macro: AMP; Wavetable: OSC; Wave seq: LANES; Sampler: TONE | 0–24 semitones | How far the pitch strip bends the sound, in semitones. |
| BIAS | Bias (wavefolder model) | Macro: MACRO | 0–100 | Third macro of the FOLD model: pushes the wave off-centre into the folder, either way of 50, which brings in the even harmonics. |
| BIAS | Key bias | Wavetable: KEYS | −100…+100 | Leans the frames between two keys towards the earlier key (down) or the later one (up). |
| BITE | Bite | TRK FX: TALK | 0–100 | Adds harmonics before the formants so quiet or pure sounds still talk. |
| BITS | Bit reduction | Sampler: TONE; TRK FX: DIRT; Insert effect: RED | 0–100 | Lowers the bit depth for digital grit. |
| BK | Chord bank | CHORD: bar | 1–4 | Which bank of sixteen chord pads (also Up / Down). |
| BPM | Tempo | SETTINGS: SETUP | 20–300 | Tempo. Shift = 0.1 steps, Func = steps of 10. |
| BUS1 | Bus 1 effect | TRK FX: TRACK | 30 types | Which effect send bus 1 carries. Shared by all tracks. |

| SHORT | FULL NAME | BELONGS TO | VALUES | WHAT IT DOES |
|---|---|---|---|---|
| BUS2 | Bus 2 effect | TRK FX: TRACK | 30 types | Which effect send bus 2 carries. Shared by all tracks. |
| CAB | Cabinet | Insert effect: AMP | 1500 Hz–10 kHz | Cutoff of the speaker-cabinet filter. |
| CC16–CC23 | General controllers 16 to 23 | MIDI track: CC | 0–127 | CC page: eight free controllers for whatever the external instrument maps them to. |
| CEIL | Ceiling | Insert effect: LMT | −3…0 dB | Highest level the limiter lets through. |
| CEIL | Master limiter ceiling | SETTINGS: MASTR | −3–0 dB | Highest level the master limiter lets through. |
| CHAN | MIDI channel | MIDI track: MIDI | 1–16 | Channel the track sends on. |
| CHAN | Envelope chain | Macro, Wavetable, Wave seq: AMP | OFF SER PAR | OFF, SER (stages follow each other) or PAR (stages run together). |
| CH.D | Chorus depth | TRK FX: MODFX | 0–100 | Amount of chorus; 0 bypasses it. |
| CHEB | Chebyshev (wavefolder model) | Macro: MACRO | 0–100 | Fourth macro of the FOLD model: harmonic-selective shaping. |
| CHOR | Chorus send (CC 93) | MIDI track: SOUND | 0–127 | Sends the chorus-level controller. |
| CH.R | Chorus rate | TRK FX: MODFX | 0.1–5 Hz | Speed of the block's chorus. |
| CL | Clear bank | CHORD: bar | four clicks | Empties the sixteen pads of this bank (an undo step). |
| CLCK | Metronome click | SETTINGS: CLICK | OFF REC PLAY | When the metronome clicks: never, while recording runs (the default), or whenever the sequencer plays. One click a beat, the first beat of each bar higher and louder. Kept between sessions, not in the project. |
| CLK | Clock source | SETTINGS: SETUP | INT EXT AUTO LINK | MIDI clock: internal, external, or automatic. LINK is listed but not available yet. |
| CLR | Clear pressure | STEP: PRES | four clicks | Removes the pad pressure recorded with the step's notes. The notes stay. |
| CLVL | Click level | SETTINGS: CLICK | 0–100 | Loudness of the click. Every turn plays one click, so it can be set with the sequencer stopped. The click is added after the master effects and MVOL: they do not change it, and it is never recorded into a resample. |
| CNT | Count-in | SETTINGS: CLICK | OFF 1BAR 2BAR | Bars the metronome counts in before a recording starts. The count-in always clicks, also with CLCK off. Nothing plays or is recorded until it ends. |
| COLR | Colour (particle noise model) | Macro: MACRO | 0–100 | Second macro of the DUST model: tone of the particles. |
| COMB | Chain combine | Macro, Wavetable, Wave seq: AMP | SUM MAX | How parallel envelope stages are combined: SUM or MAX. |
| COND | Trig condition | STEP: TRIG | ALL PROB A:B FILL !FIL 1ST !1ST PRE !PRE | When the step plays (Appendix C). |
| CR | Crop | WAVE: REC | four clicks | Trims the sample to its start and end markers. |
| CT | Category | PRESET: bar | list | Filters the list by category. |
| CURV | Shaping curve | Insert effect: WSH | SOFT HARD FOLD ASYM | SOFT, HARD, FOLD or ASYM transfer curve. |
| CURV | Envelope curve | Synth: F ENV, AMP; FM, Macro, Wavetable, Wave seq: F ENV | −100…+100 | Shape of the envelope segments, from logarithmic through linear to exponential. |

| SHORT | FULL NAME | BELONGS TO | VALUES | WHAT IT DOES |
|---|---|---|---|---|
| CUT | Cutoff (CC 74) | MIDI track: MIDI | 0–127 | Sends the filter cutoff controller to the external instrument. |
| CUT | Cutoff | Synth, FM, Macro, Wavetable, Wave seq, Sampler: FILTR; Insert effect: FLT | varies | Filter cutoff frequency. |
| CUTB | Filter B cutoff | Synth: LFO DEST | destination | The cutoff of filter B alone, up to four octaves either way. |
| DAMP | Damping | Insert effect: RVB, HALL, SPRG, SHIM | 0–100 | How quickly high frequencies die away in the tail. |
| DAMP | Damping (string models) | Macro: MACRO | 0–100 | Macro of the PLCK and BOW models: string damping. |
| DAMP | Damping | TRK FX: DIRT | 0–100 | Darkens the DIRT block's feedback. |
| DC | Decay | DRUMS: bar | 0.01–4 s | DEC of the selected drum pad. |
| DCAY | Decay time | Insert effect: RVB, HALL, SPRG, SHIM | varies | Length of the reverb tail. |
| DEC | Decay | Drums: DR | 0.01–4 s | Length of the drum voice. A drum rings for its DEC however short the pad press or the step; SUB sounds while held and fades over DEC afterwards. |
| DEC | Decay (CC 75) | MIDI track: SOUND | 0–127 | Sends the decay-time controller. |
| DEC | Decay | Synth: F ENV, AMP; FM: OP1–OP6, F ENV; Macro, Wavetable, Wave seq: AMP, F ENV; Sampler: AMP | varies | Time the envelope takes to fall to the sustain level. |
| DECn | Envelope decay | Synth, FM: LFO DEST | destination | FM, DEC1 to DEC6: the decay of operator 1 to 6. Synth, DEC2 to DEC5: the decay of an extra envelope (MENV). |
| DENS | Density | ARP: VARY | 0–100 | How many steps sound. |
| DENS | Density (particle noise model) | Macro: MACRO | 0–100 | First macro of the DUST model: particle density. |
| DENS | Grain density | Sampler: GRAIN | 1–200 Hz | Grains started per second. |
| DEST | Pressure destination | ARP: ACFG | RATE OCT GATE | What pad pressure does to the arp. |
| DEST | Route destination | AT ROUTE: top row | per engine | What pad pressure moves, for routes 1 to 4 of this track. |
| DEST | LFO destination | FM, Macro, Wavetable, Wave seq, Sampler, Drums: LFO1 / LFO2 | varies | What the LFO moves; the list depends on the engine (Appendix C). |
| DET2 | Oscillator 2 detune | Synth: OSC | −50…50 cents | Fine detune of oscillator 2 in cents, for beating and width. |
| DETn | Operator detune | FM: LFO DEST | destination | DET1 to DET6: the DETN cell of operator 1 to 6. |
| DETN | Operator detune | FM: OP1–OP6 | −50…50 cents | Fine detune of this operator in cents. |
| DETN | Detune (FM model) | Macro: MACRO | 0–100 | Third macro of the FM model: detunes the modulator up to a semitone either way of 50; the tone beats and turns inharmonic. |
| DICE | Dice | ARP: VARY | 0–100 | Re-rolls the random choices behind the variations. |
| DIR | Strum direction | ARP: PLAY | UP DOWN ALT | Direction of the strum. |

| SHORT | FULL NAME | BELONGS TO | VALUES | WHAT IT DOES |
|---|---|---|---|---|
| DIR | Scan direction | Wavetable: SCAN | FWD REV | FWD runs from POS to the end of the table; REV from the end back to POS. |
| DISP | Dispersion | Insert effect: SPRG | 0–100 | The spring's characteristic chirp. |
| DIV | LFO division | FM, Macro, Wavetable, Wave seq, Sampler, Drums: LFO1 / LFO2 | 17 values (App. C) | The tempo division used while SYNC is on: 8 bars to 1/32, with dotted and triplet values. |
| DL | Delete | PRESET, SONG: bar | four clicks | Deletes the user preset (PRESET) or the selected row (SONG). |
| DLY | LFO delay | Synth: MOD2 | seconds | Wait after note-on before the LFO starts. |
| DPTH | Pressure depth | ARP: ACFG / ROLL | 0–100 | How far pressure moves it. |
| DPTH | Route depth | AT ROUTE: bottom row | −100…+100 | How far; drums 0 to 100. |
| DPTH | Depth | FM, Macro, Wavetable, Wave seq, Sampler, Drums: LFO1 / LFO2; Insert effect: CHR, FLG, PHS, TRM, ENS, PAN, RING, VIB | varies | Amount of modulation. On LFOs negative values invert the shape. |
| DPTH | Vowel LFO depth | TRK FX: TALK | 0–100 | How far the vowel moves by itself. |
| DPTn | LFO depth | Synth: LFO DEST | destination | DPT1 to DPT4: the depth of LFO 1 to 4, from nothing to double. |
| DRAT | Drift rate | Synth: ANALG | 0.05–5 Hz | Speed of the pitch drift. |
| DRFT | Drift amount | Synth: ANALG | 0–100 | Slow random wander of each oscillator's pitch. |
| DRIV | Drive | Drums: DR | 0–100 | Saturation of this drum voice, on every type. The snares also shift from body to noise with it, CLAP adds claps, SHKR rattles deeper. |
| DRIV | Drive | Sampler: TONE | 0–100 | Saturation of the sampler voice. |
| DRIV | Filter drive | Synth, FM, Macro, Wavetable, Wave seq: FILTR | 0–100 | Input drive into the filter. Each type overdrives in its own way; the SVF soft-clips its input, and on COMB it darkens the feedback. |
| DRIV | Mixer drive | Synth: OSC | 0–100 | Overdrives the oscillator mix before the filter. |
| DRIV | Drive | TRK FX: DIRT; Insert effect: DRV, WSH | 0–100 | Amount of distortion. |
| DRUM | Drum speed | Insert effect: ROT | 0.5–2 Hz | Fast-speed rotation of the bass rotor. |
| DRVB | Filter B drive | Synth: FLT B | 0–100 | Input drive into filter B. |
| DST1 | Strip route 1 destination | FM, Macro, Wavetable, Wave seq, Sampler, Drums: STRIP | varies | STRIP page: what the mod strip moves on route 1. The same list as the LFOs' DEST (Appendix C); from the strip, LRAT and LDPT act on both LFOs. Func + turn goes from group to group. |
| DST2 | Strip route 2 destination | FM, Macro, Wavetable, Wave seq, Sampler, Drums: STRIP | varies | STRIP page: what the mod strip moves on route 2. The same list as the LFOs' DEST (Appendix C); from the strip, LRAT and LDPT act on both LFOs. Func + turn goes from group to group. |
| DST3 | Strip route 3 destination | FM, Macro, Wavetable, Wave seq, Sampler, Drums: STRIP | varies | STRIP page: what the mod strip moves on route 3. The same list as the LFOs' DEST (Appendix C); from the strip, LRAT and LDPT act on both LFOs. Func + turn goes from group to group. |
| DV | Drive | DRUMS: bar | 0–100 | DRIV of the selected drum pad. |
| ED | Edit | SCALE: bar | OFF / ON | Edit mode: tap a pad, then a key assigns its note. |

| SHORT | FULL NAME | BELONGS TO | VALUES | WHAT IT DOES |
|---|---|---|---|---|
| EDPT | Extra envelope depth | Synth: LFO DEST | destination | The DPTH cell of the MENV page. |
| EN | Engine filter | PRESET: bar | THIS / ANY | Show presets for this track's engine only, or all. |
| EN | End | WAVE: WAVE | position | End marker (END). |
| END | End | Sampler: SAMPL | 0–100 | Where playback ends. |
| ENG | Engine | TRACKS: ENG | 8 engines | Knob N chooses a pending engine for track N; that track's top pad commits it. |
| ENV | Envelope amount | Synth, FM, Macro, Wavetable, Wave seq: FILTR | −100…+100 | How far the filter envelope moves the cutoff; negative closes the filter. Up to 7 octaves. |
| ENV | Envelope to vowel | TRK FX: TALK | −100…+100 | How far the playing level pushes the vowel; negative pushes it the other way. |
| ENVF | Envelope to filter | Sampler: FILTR | −100…+100 | How far the amp envelope moves the cutoff; negative closes it. |
| ENVP | Envelope to position | Wavetable: WAVE | −100…+100 | The amp envelope moves the table position. |
| ENVS | Envelope count | Synth: MENV | 0–4 | How many extra envelopes each voice runs. |
| EVAR | Envelope variation | Synth: ANALG | 0–100 | Random change of envelope times per note. |
| EVEL | Velocity to envelope amount | FM, Macro, Wavetable, Wave seq: F ENV | 0–100 | Harder notes increase the filter envelope's reach. |
| EVOD | Even / odd (harmonic model) | Macro: MACRO | 0–100 | Second macro of the HARM model: balance of even and odd harmonics. |
| EVOL | Evolution | ARP: MUTA | 0–99 | Generations mutated so far; anticlockwise returns to the ancestor (an undo step). |
| EXPR | Expression (CC 11) | MIDI track: MIDI | 0–127 | Sends the expression controller. |
| FADE | LFO fade-in | FM, Macro, Wavetable, Wave seq, Sampler, Drums: LFO1 / LFO2 | 0–5 s | Time the LFO takes to reach full depth after a note starts. |
| FADE | Fade-in | Insert effect: VIB | 0–2 s | Time the vibrato takes to come in. |
| FADE | Slice fade | Sampler: SLICE | 0–0.1 s | Fades the end of each slice to avoid clicks. |
| FATK | Filter envelope attack | Synth, FM, Macro, Wavetable, Wave seq: LFO DEST | destination | The ATK cell of the F ENV page. |
| FDBK | Operator feedback | FM: OP1–OP6 | 0–100 | The operator modulates itself; adds bite, then noise. |
| FDBK | Feedback | Insert effect: FLG, PHS, DLY, RVSE, TAPE, LCR, MTAP | varies | How much of the effect's output is fed back into it. |
| FDBK | Feedback (plucked string and FM models) | Macro: MACRO | 0–100 | Third macro of the PLCK model: string feedback (sustain). Fourth macro of the FM model: the modulator feeds back into itself; bite, then noise. |
| FDBK | Feedback | TRK FX: DIRT | 0–100 | Tuned feedback around the distortion; howls at high settings. |
| FDBn | Operator feedback | FM: LFO DEST | destination | FDB1 to FDB6: the FDBK cell of operator 1 to 6. |
| FDEC | Filter envelope decay | Synth, FM, Macro, Wavetable, Wave seq: LFO DEST | destination | The DEC cell of the F ENV page. |
| FDRV | Filter drive | Sampler: FILTR | 0–100 | Input drive into the sampler's filter. |
| FFM | Filter FM | Synth: FILTR | 0–100 | Oscillator 2 modulates the cutoff at audio rate. |
| FILT | Smoothing filter | Insert effect: RED | 0–100 | Softens the reducer's aliasing. |

| SHORT | FULL NAME | BELONGS TO | VALUES | WHAT IT DOES |
|---|---|---|---|---|
| FINE | Fine tune | Synth: OSC3; Macro: MACRO; Wavetable: OSC; Wave seq: SEQ; Sampler: TONE | varies | Fine pitch of the whole sound. |
| FL.D | Flanger depth | TRK FX: MODFX | 0–100 | Amount of flanger; 0 bypasses it. |
| FL.R | Flanger rate | TRK FX: MODFX | 0.1–5 Hz | Speed of the block's flanger. |
| FMOR | Filter morph | Macro: LFO DEST | destination | The MORP cell of the FILTR page (the macro oscillator's own MORP is its fourth macro). |
| FOLD | Fold (wavefolder model) | Macro: MACRO | 0–100 | First macro of the FOLD model: fold amount. |
| FREL | Filter envelope release | Synth, FM, Macro, Wavetable, Wave seq: LFO DEST | destination | The REL cell of the F ENV page. |
| FREQ | Carrier frequency | Insert effect: RING | 20 Hz–5 kHz | Frequency the signal is multiplied with. |
| FREQ | Wah frequency | Insert effect: WAH | 150 Hz–2.5 kHz | Centre of the wah's sweep. |
| FREZ | Freeze | ARP: PLAY | OFF / ON | Holds the current note order while you change keys. |
| FRMS | Frames | Wavetable: SCAN | 64 128 256 A64 A128 A256 | How a sample is cut into the USER table: 64, 128 or 256 equal pieces, or A64 / A128 / A256, aligned to the sound's pitch period so scanning is smooth and plays in tune. |
| FSUS | Filter envelope sustain | Synth, FM, Macro, Wavetable, Wave seq: LFO DEST | destination | The SUS cell of the F ENV page. |
| FVAR | Filter variation | Synth: ANALG | 0–100 | Random cutoff offset per note. |
| FVEL | Velocity to cutoff | Synth, FM, Macro, Wavetable, Wave seq: LFO DEST | destination | The VEL cell of the F ENV page. |
| GAIN | Gain | Insert effect: AMP | 0–100 | Amp preamp gain. |
| GAP | Minimum gap | Sampler: SLICE | 0.01–1 s | Shortest time allowed between two detected hits; raise it to merge flams and rolls into one slice. |
| GATE | Gate | ARP: ARP / ROLL | 0–100 | Length of each arp note (or roll repeat) as a share of its step. |
| GATE | Arp gate lock | TRK FX: ARPLK | 0–100 | Per-step lock of the arpeggiator's gate length. |
| GLID | Glide | Synth: VOICE; Wavetable: OSC | 0–2 s | Portamento time between notes. |
| GLMD | Glide mode | Synth: VOICE | ALL LEGA | ALL: every note glides. LEGA: only overlapping notes glide. |
| GPOS | Grain position | Sampler: LFO DEST | destination | GRAN mode: where new grains start, around the scan position. |
| GVAR | Gate variation | ARP: VARY | 0–100 | Spread of gate lengths. |
| HARM | Harmonics (virtual analog model) | Macro: MACRO | 0–100 | Third macro of the VA model. 50 is the saws as they are; below, each fades into a sine; above, a saw an octave up is mixed in. |
| HARM | Highest harmonic | Wavetable: KEYS | 1–127 | The highest harmonic kept of every key of the KEYS table, 1 to 127. |
| HCUT | High cut | Insert effect: RVSE | 500 Hz–20 kHz | Darkens the reversed echoes. |
| HELP | Help cards | SETTINGS: SETUP | OFF 0.35S INST | When holding Shift or Func alone shows its help card: never, after 0.35 seconds (the default), or at once. Kept between sessions. |
| HIGH | High gain | Insert effect: EQ, GEQ | varies | Boost or cut of the high band. |

| SHORT | FULL NAME | BELONGS TO | VALUES | WHAT IT DOES |
|---|---|---|---|---|
| HIGH | Master EQ high | SETTINGS: MASTR | ±18 dB | High band of the master EQ. |
| HMID | High-mid gain | Insert effect: GEQ | −12…12 dB | Boost or cut of the high-mid band. |
| HOLD | Kick hold | Drums: KICK | 0–0.3 s | Kick machine: time at full level before the decay starts. |
| HORN | Horn speed | Insert effect: ROT | 5–10 Hz | Fast-speed rotation of the treble horn. |
| HUMN | Humanize | ARP: PLAY / ROL2 | 0–100 | Random variation of timing and velocity. |
| IN | Insert row | SONG: bar | turn | Inserts a row. |
| INDX | Index (FM model) | Macro: MACRO | 0–100 | First macro of the FM model: modulation depth. |
| INTP | Interpolation | Wavetable: WAVE | LIN SPEC STEP VINT CLSC | How the table is read: LIN smooth, SPEC spectral, STEP stepped, VINT vintage (gritty), CLSC classic: as an early digital wavetable synth read its tables (64 positions, waves of 128 samples and 8 bits, nothing smoothed or band-limited), which gives high notes a glassy, aliased top. |
| INTV | Mutate interval | ARP: MUTA | 1–16 | Bars between mutations. |
| JIT | Jitter | Sampler: STRCH | 0–100 | Randomises where each cycle or grain starts; smooths the buzz of CYCL, widens the blur of SMER. |
| KEY | Key follow | Sampler: TONE | OFF ON | On: pitch follows the note. Off: every note plays the sample at its original pitch. |
| KEY | Key tracking | Synth: F ENV | −100…+100 | How far the cutoff follows the played note, -100 to +100: at +100 it rises an octave per octave played (on the sampler a tenth of its range), at 0 it stays put, below 0 it falls as the notes rise. Middle C is the centre. |
| KFRM | Key table frames | Wavetable: KEYS | 8 16 32 64 128 256 | How many frames the KEYS table has: 8 to 256. Few frames and INTP on STEP or CLSC make the steps of a scan audible. |
| KM | Key mapping | SCALE: bar | root / number | Keys pick a page by its root or by its number. |
| KNCK | Knock | Drums: KICK | 0–100 | Kick machine: level of the short tick at the very start of the drum. |
| KS | Keys select pages | SCALE: bar | OFF / latched / momentary | Keyboard keys choose the scale page. |
| KTON | Knock tone | Drums: KICK | 0–100 | Kick machine: from a dull thud of about 12 ms to a bright tick of about 1.5 ms. |
| KTRK | Key tracking | FM, Macro, Wavetable, Wave seq, Sampler: FILTR | −100…+100 | How far the cutoff follows the played note, -100 to +100: at +100 it rises an octave per octave played (on the sampler a tenth of its range), at 0 it stays put, below 0 it falls as the notes rise. Middle C is the centre. |
| KY | Key frame | WAVE: WAVE (wavetable track) | 1–8 | Which key frame of the KEYS table the page shows and edits. The top-row pads choose it too. |
| L1–L16 | Step level | Wave seq: LVL A / B | 0–100 | LVL A / LVL B pages: level of steps 1 to 16. |
| >LCK | Pressure to locks | STEP: PRES | four clicks | Turns the step's recorded pressure into parameter locks on the steps the note spans, through the track's pressure routes as they are now, then removes the recording. |
| LCKS | Locks | STEP: TRIG | count | Number of parameter locks on the step; turning it anticlockwise clears them (an undo step). |

| SHORT | FULL NAME | BELONGS TO | VALUES | WHAT IT DOES |
|---|---|---|---|---|
| LD | Load | PRESET: bar | four clicks | Keeps the highlighted preset as the track's sound for good. |
| LDPT | Other LFO's depth | FM, Macro, Wavetable, Wave seq, Sampler, Drums: LFO DEST | destination | The depth of the other LFO, from nothing to double. |
| LE | Loop end | WAVE: WAVE | position | Loop end marker (LPEN). |
| LEGA | Legato | Synth: VOICE | OFF ON | On: overlapping notes do not retrigger the envelopes. |
| LEN | Note length | STEP: TRIG | 0.25–16 steps | How long the step's note is held. |
| LEN | Stretch length | Sampler: STRCH | 10 values (App. C) | With SYNC on, how long the loop lasts: 1/16 of a bar to 32 bars. |
| LEN | Track length | TRK FX, STEP: TRACK / bar | 1–1024 | Number of steps in this track's pattern; every track can differ (polymeter). STEP shows it as LN. |
| LEN | Sequence length | Wave seq: SEQ | 1–16 | Number of steps in the wave sequence. |
| LEN | Scan length | Wavetable: SCAN | 1/16 1/8 1/4 1/2 1BAR 2BAR 4BAR 8BAR 16BR | With SYNC on, how long one pass lasts: 1/16 to 16 bars. |
| LEVL | Level | Drums: DR | 0–100 | Level of this drum voice. |
| LEVL | Operator level | FM: OP1–OP6 | 0–100 | Output level of this operator: loudness for a carrier, modulation depth for a modulator. |
| LEVL | Output level | Insert effect: AMP | −24…12 dB | Output level of the amp effect in dB. |
| LEVL | Level | Synth: AMP; FM: FMMIX; Macro: MACRO; Wavetable: WAVE; Wave seq: SEQ; Sampler: TONE | 0–100 | Output level of the engine, before the track's mixer volume. |
| LFO | LFO select | Synth: MOD / MOD2 | LFO1–LFO4 | Which LFO the page edits. |
| LFOS | LFO count | Synth: MOD | 1–4 | How many LFOs each voice runs. |
| LK | Scale lock | SCALE: bar | OFF / ON | Forces keys and arps into the scale (also the Scale button). |
| LLEN | Level lane length | Wave seq: LANES | 1–16 | Length of the level lane. Lanes of different lengths drift against each other. |
| LMID | Low-mid gain | Insert effect: GEQ | −12…12 dB | Boost or cut of the low-mid band. |
| LN | Track length | STEP: bar | 1–1024 | Length of the selected track in steps. |
| LN | Length | WAVE: REC | free, 1 beat … 8 bars | Length of the take. |
| LOAD | Load project | SETTINGS: PROJ | four clicks | Loads the slot. Shows OK or FAIL. |
| LOCK | Macro / lock interaction | ARP: ACFG | ADD MACRO LOCK MULT | How variation macros and step locks combine. |
| LOOK | Look-ahead | Insert effect: LMT | 1–5 ms | How far ahead the limiter looks, in milliseconds. |
| LOOP | Loop mode | Wave seq: SEQ | FWD REV PING RAND ONCE | FWD, REV, PING, RAND or ONCE. |
| LOW | Low gain | Insert effect: EQ, GEQ | varies | Boost or cut of the low band. |
| LOW | Master EQ low | SETTINGS: MASTR | ±18 dB | Low band of the master EQ. |
| LP | Loop | SONG: bar | OFF / ON | Loops the song at the end. |
| LPEN | Loop end | Sampler: SAMPL | 0–100 | End of the looped region. |
| LPST | Loop start | Sampler: SAMPL | 0–100 | Start of the looped region. |

| SHORT | FULL NAME | BELONGS TO | VALUES | WHAT IT DOES |
|---|---|---|---|---|
| LRAT | Other LFO's rate | FM, Macro, Wavetable, Wave seq, Sampler, Drums: LFO DEST | destination | The rate of the other LFO, up to two octaves either way. |
| LS | Loop start | WAVE: WAVE | position | Loop start marker (LPST). |
| LTCH | Latch | ARP: ARP / ROL2 | OFF / ON | ARP: the arp keeps playing after the keys are released (also Shift + Arp). ROL2: a pad press starts a roll that runs until pressed again. |
| LV | Level | DRUMS: bar | 0–100 | LEVL of the selected drum pad. |
| LVL1–LVL6 | Operator level | FM: FMMIX | 0–100 | FMMIX page: output level of operators 1 to 6, one knob each. |
| LVL3 | Oscillator 3 level | Synth: OSC3 | 0–100 | Level of oscillator 3 (sub). |
| LVLn | Oscillator level | Synth: LFO DEST | destination | LVL1 to LVL3: the level of one oscillator, from silence to double. |
| MAKE | Make-up gain | Insert effect: CMP | 0–30 dB | Gain after compression. |
| MAKE | Master make-up gain | SETTINGS: MASTR | 0–30 dB | Gain after the master compressor. |
| MAX | Maximum rate | ARP: ROLL | division or Hz | Repeat rate at full pressure. |
| MD | Mode | SCALE: bar | RL / AB | Relative (pads follow the root) or absolute. |
| MD | Play mode | WAVE: WAVE | FWD … GRAN | The sampler's MODE. |
| MDPT | Motion depth | Wavetable: WAVE | 0–100 | Depth of the built-in position sweep. |
| MFRQ | Mid frequency | Insert effect: EQ | 100 Hz–8 kHz | Centre frequency of the mid band. |
| MFRQ | Master EQ mid frequency | SETTINGS: MASTR | 100 Hz–8 kHz | Centre of the mid band. |
| MICR | Micro-timing | STEP: TRIG | −50…+50 % | Moves the step earlier or later by up to half a step. |
| MID | Mid gain | Insert effect: EQ | −18…18 dB | Boost or cut of the mid band. |
| MID | Master EQ mid | SETTINGS: MASTR | ±18 dB | Mid band of the master EQ. |
| MIX | Mix (particle noise model) | Macro: MACRO | 0–100 | Fourth macro of the DUST model: balance of the noise particles against the tone. |
| MIX | Oscillator mix | Synth: OSC | 0–100 | Balance between oscillator 1 and oscillator 2. |
| MIX | Dry / wet mix | TRK FX: DIRT, TALK; Insert effect: CHR, DRV, FLG, DLY, RVB, ENS, RED, RVSE, RING, TAPE, LCR, MTAP | 0–100 | Balance between the untreated sound (0) and the effect (100). |
| MOD | Modulation (CC 1) | MIDI track: MIDI | 0–127 | Sends the mod-wheel controller. |
| MODB | Filter B mode | Synth: FLT B | --- | Response of the second filter. |
| MODE | Arp mode | ARP: ARP | 15 modes | Order the held notes play in (Appendix C). |
| MODE | Delay mode | Insert effect: DLY | STER PING TAPE | STER stereo, PING ping-pong, TAPE with tape-style tone. |
| MODE | Shimmer interval | Insert effect: SHIM | OCT 5TH O+5 | Pitch shift fed into the reverb: octave, fifth, or both. |
| MODE | Play mode | Sampler: SAMPL | FWD REV 1SHT GATE LOOP PING GRAN | FWD forward, REV reverse, 1SHT one-shot, GATE while held, LOOP, PING back-and-forth loop, GRAN granular. |
| MODE | Slice mode | Sampler: SLICE | GRID TRNS | GRID: equal slices. TRNS: one slice per detected hit (transient). |

| SHORT | FULL NAME | BELONGS TO | VALUES | WHAT IT DOES |
|---|---|---|---|---|
| MODE | Filter mode | Synth, FM, Macro, Wavetable, Wave seq, Sampler: FILTR; Insert effect: FLT | LP BP HP NTCH | Response of the chosen filter type (low-pass, band-pass, high-pass, notch, slopes); the list depends on TYPE. |
| MODE | Voice mode | Synth: VOICE | MONO DUO POLY UNI | MONO, DUO (two voices), POLY, or UNI (all voices stacked on one note). |
| MODE | Scan mode | Wavetable: SCAN | OFF RUN LOOP PING STEP RAND | Moves the table position by itself over a set time: OFF, RUN (once, then holds), LOOP, PING (there and back), STEP (in audible steps), RAND (random frames). |
| MODL | Model | Macro: MACRO | VA FM FOLD FORM HARM PLCK DUST BOW | Synthesis model: VA, FM, FOLD, FORM, HARM, PLCK, DUST, BOW. Knobs 2 to 5 are renamed to suit the model. |
| MORP | Morph | Macro: MACRO, FILTR | 0–100 | Fourth macro of the VA, FORM and PLCK models. VA: saw to square. FORM: voice to whisper (noise takes over from the pulse train). PLCK: string to drum (the pitch dissolves into a burst). |
| MORP | Filter morph | Synth, FM, Wavetable, Wave seq: FILTR | 0–100 | On the SEM type in MRPH mode: sweeps low-pass through notch to high-pass. Anywhere else the knob says SEM MRPH ONLY. |
| MSPD | Motion speed | Wavetable: WAVE | 0–10 Hz | Speed of the built-in sweep of the table position. |
| MUTA | Mutate | ARP: MUTA | OFF / ON | Switches mutation on: the arp pattern changes by itself over time. SPCE on the same page sets how much. |
| MVOL | Master volume | SETTINGS: SETUP | 0–100 | Overall output level. |
| nA | Oscillator source, first cell | Synth: LFO DEST | destination | 1A, 2A, 3A: the first SRC cell of oscillator 1, 2 or 3 (table position, FM index, formant frequency or harmonic tilt, by what the slot plays). |
| nB | Oscillator source, second cell | Synth: LFO DEST | destination | 1B, 2B, 3B: the second SRC cell of oscillator 1, 2 or 3 (table, FM ratio, formant width or odd / even balance); tables and ratios change in steps. |
| NCOL | Noise colour | Synth: OSC3 | 0–100 | Tone of the noise, from dark to bright. |
| NEW | New project | SETTINGS: PROJ | four clicks | Starts an empty project. |
| nFRM | Formant frequency | PARAM: SRC | 0–100 | Synth, slot n set to FORM: the resonant peak, from 1 to 16 times the note's pitch. |
| nIDX | FM index | PARAM: SRC | 0–100 | Synth, slot n set to FM: how hard the modulator pushes the carrier. |
| NM | Normalise | WAVE: REC | OFF / ON | Raises the take to full level. |
| nODD | Odd harmonics | PARAM: SRC | 0–100 | Synth, slot n set to HARM: from all harmonics to odd ones only. |
| NOIS | Noise level | Synth: OSC3 | 0–100 | Level of the noise source. |
| NORM | Normalise | Sampler: SAMPL | OFF ON | On: the sample's loudest point is brought to full level. |
| NORM | Frame level | Wavetable: KEYS | OFF PEAK RMS | Level of the frames of the KEYS table. OFF: as computed. PEAK: every frame at full scale. RMS: every frame equally loud. |
| NOTE | Step note | STEP: TRIG | C-1 … G9 | Pitch of the step; a chord moves together. |
| nPOS | Wavetable position | PARAM: SRC | 0–100 | Synth, slot n set to WT: where in the table the oscillator reads. |

| SHORT | FULL NAME | BELONGS TO | VALUES | WHAT IT DOES |
|---|---|---|---|---|
| nRAT | FM ratio | PARAM: SRC | 1–12 | Synth, slot n set to FM: the modulator's pitch as a whole multiple of the note. |
| NT | Note | SCALE: bar (edit) | note | Edit mode: the selected pad's note. |
| nTBL | Wavetable table | PARAM: SRC | SAW … DIGI, VOWL … COMB | Synth, slot n set to WT: which of the wavetable engine's 23 built-in tables (the first nine, then the fourteen newer ones). |
| nTLT | Harmonic tilt | PARAM: SRC | 0–100 | Synth, slot n set to HARM: from all harmonics equally loud (bright) to steeply falling (dark). |
| NUM | Number (harmonic model) | Macro: MACRO | 0–100 | Third macro of the HARM model: number of harmonics. |
| nWID | Formant width | PARAM: SRC | 0–100 | Synth, slot n set to FORM: how broad the peak is. |
| OC | Octave | CHORD: bar | −3 to +3 | Moves every chord pad by octaves. |
| OC | Octave | SCALE: bar | octaves | Octave of the pads. |
| OCT | Octave range | ARP: ARP | 1–8 | Octaves the arp spans. |
| OCT | Octave | Wavetable: OSC | −4…4 | Octave shift of the oscillator. |
| OFF | No destination | all engines: LFO DEST | destination | The LFO does nothing and costs nothing. |
| OFSB | Filter B offset | Synth: FLT B | −48…48 semitones | Cutoff of filter B relative to filter A, in semitones. |
| ON | Insert on | TRK FX: FX | OFF / ON | Switches the slot on or off. Effects with a TIME or RATE show SYNC here instead. |
| OUT | Output gain | Insert effect: TRNS | −12…12 dB | Output level in dB. |
| P1—P16 | Step pitch | Wave seq: PIT A / B | −24…24 semitones | PIT A / PIT B pages: pitch offset of steps 1 to 16, one knob per step. |
| PA | Pattern | SONG: bar | 1–16 | Pattern this row plays. |
| PAN | Pan (CC 10) | MIDI track: MIDI | 0–127 | Sends the pan controller. |
| PAN | Pan | Sampler: AMP; Drums: DR | varies | Left / right position. |
| PAN | Track pan | TRACKS: PAN | L–R | Mixer page: knob N is the pan of track N. |
| >PAT | Write to pattern | ARP: MUTA | four clicks | Writes the arp's current steps into the track's pattern (an undo step). |
| PAT | Pattern | SETTINGS: SETUP | 1–16 | Selects the pattern (same as > + pad). |
| PD | Pad | SCALE: bar (edit) | 1–16 | Edit mode: which pad is being edited. |
| PDLY | Pre-delay | Insert effect: HALL | 0–100 ms | Gap before the reverb starts. |
| PG | Scale page | SCALE: bar | 12 factory + user | Which pad layout (scale) the pads play. |
| PHAS | Stereo phase | Insert effect: PAN | 0–100 | Phase offset between left and right of the auto-pan. |
| PHAS | Random phase | Synth: ANALG | OFF ON | On: oscillators start at a random point in their cycle on each note. |
| PH.D | Phaser depth | TRK FX: MODFX | 0–100 | Amount of phaser; 0 bypasses it. |
| PH.R | Phaser rate | TRK FX: MODFX | 0.1–5 Hz | Speed of the block's phaser. |
| PICK | Pick (plucked string model) | Macro: MACRO | 0–100 | Second macro of the PLCK model: pick position. |
| PITn | Oscillator 2 pitch | Synth: LFO DEST | destination | PIT2: the pitch of oscillator 2 alone, up to an octave either way. |
| PJIT | Pitch jitter | Sampler: GRAIN | 0–100 | Random pitch per grain, up to an octave either way. |

| SHORT | FULL NAME | BELONGS TO | VALUES | WHAT IT DOES |
|---|---|---|---|---|
| PL | Key place | WAVE: WAVE (wavetable track) | 0–100 | Where along the table the key sits (PLCn). |
| PLAY | Slice play | Sampler: SLICE | ONE KEYS NEXT RAND | ONE: the slice chosen by SLIC. KEYS: each key plays its own slice, upward from ROOT. NEXT: each new note plays the next slice. RAND: a random slice per note. |
| PLC1–PLC8 | Key place | Wavetable: K PLC | 0–100 | K PLC page: where along the table key frame 1 to 8 sits. Only order and spacing count: the earliest key in use is the first frame, the latest the last. |
| PLEN | Pitch lane length | Wave seq: LANES | 1–16 | Length of the pitch lane. |
| PMOD | Poly-mod | Synth: F ENV | −100…+100 | Filter envelope to oscillator 2 pitch, up to two octaves either way. |
| PN | Pan | DRUMS: bar | L–R | PAN of the selected drum pad. |
| PO | Key position | WAVE: WAVE (wavetable track) | position | Where in the sample the key's cycle is taken (POSn): the line on the waveform. |
| POS | Position (bowed string model) | Macro: MACRO | 0–100 | Second macro of the BOW model: bow position. |
| POS | Table position | Wavetable: WAVE | 0–100 | Where in the wavetable the oscillator reads; sweeping it morphs the tone. |
| POS1–POS8 | Key position | Wavetable: K POS | 0–100 | K POS page: where in its sample key frame 1 to 8 is taken. One cycle is cut there (one period where the sample has a pitch) and plays at the pitch of the keys. |
| PRES | Pressure (bowed string model) | Macro: MACRO | 0–100 | First macro of the BOW model: bow pressure, from a light, glassy bow to a pressed, rough one. |
| PRES | Recorded pressure amount | STEP: PRES | 0–200 % | How much of the pad pressure recorded with the step's notes is played back. 0 plays none and keeps the recording; hold > and turn for 100. |
| PRNG | Pitch range | Wave seq: SEQ | 0–2× | Scales how far the pitch lane reaches (0 = flat, 2 = double). |
| PROB | Repeat probability | ARP: ROL2 | 0–100 | Chance each repeat sounds. |
| PROB | Probability | STEP: TRIG | 0–100 % | Chance the step plays, with COND = PROB. |
| PROG | Program | MIDI track: MIDI | 0–127 | Sends a program change. |
| PS | Preset | PRESET: bar | list | Scrolls the preset list. |
| PTCH | Pitch | Sampler: TONE | −24…24 semitones | Transposition of the sample in semitones. |
| PTCH | Pitch | all engines: LFO DEST | destination | Pitch of the whole voice, up to an octave either way; the lower half of DPTH is vibrato territory. |
| P>V | Pressure to velocity | ARP: ROLL | 0–100 | How much pressure sets each repeat's velocity. |
| PW | Pulse width / shape | Synth: OSC | 5–95 | Shapes both oscillators: 50 is the plain wave, and the two directions are two different sounds. Below 50 thins or sharpens: SQR and PLS narrow to a thin pulse, SIN and TRI are wavefolded, SAW gains a second, shifted saw. Above 50 thickens: SQR and PLS gain a square an octave up, SIN and TRI are pushed towards a square, SAW hollows into a square. |
| Q | Q (formant model) | Macro: MACRO | 0–100 | Third macro of the FORM model: formant sharpness. |

| SHORT | FULL NAME | BELONGS TO | VALUES | WHAT IT DOES |
|---|---|---|---|---|
| QN | Playback quantize | STEP: bar | 0–100 | Pulls recorded notes toward the grid on playback; the recording itself is never changed. |
| QN | Start quantize | WAVE: REC | NOW / BAR | Start at once, or on the next bar. |
| QNT | Quantize | SETTINGS: SETUP | 0–100 | Global playback quantize. |
| RAND | Random start | Sampler: AMP | 0–100 | Each note starts at a random point up to this far into the sample. |
| RAT1–RAT6 | Operator ratio | FM: RATIO | 0.5–16 ratio | RATIO page: frequency ratio of operators 1 to 6, one knob each. A turn steps through the harmonics (0.5, 1, 2, 3 ... 16); Shift + turn moves by 0.01 for in-between ratios; Func + turn by an octave. |
| RATE | Arp rate | ARP: ARP | 1/1–1/64T | Note value of the arp's steps. |
| RATE | Rate | FM, Macro, Wavetable, Sampler, Drums: LFO1 / LFO2; Insert effect: CHR, FLG, PHS, TRM, ENS, PAN, RING, VIB, WAH | varies | Speed of the LFO or effect modulation. With SYNC on it picks a tempo division instead. |
| RATE | Sample-rate reduction | Sampler: TONE | 0–100 | Lowers the playback sample rate for aliasing grit. |
| RATE | Sample-rate reduction | TRK FX: DIRT; Insert effect: RED | 0–100 | Lowers the sample rate for aliasing grit. |
| RATE | Vowel LFO rate | TRK FX: TALK | 0.05–12 Hz | Speed of the vowel movement. |
| RATE | Rate | Wave seq: SEQ, LFO1 / LFO2 | varies | Speed of the wave sequence relative to the tempo. |
| RATI | Operator ratio | FM: OP1–OP6 | 0.5–16 ratio | Frequency of this operator as a multiple of the note. |
| RATI | Compression ratio | Insert effect: CMP, DUCK | 1–20× | How strongly level above the threshold is reduced. |
| RATI | Ratio (FM model) | Macro: MACRO | 0–100 | Second macro of the FM model: modulator ratio. |
| RATI | Master compressor ratio | SETTINGS: MASTR | 1–20 | Strength of the master compression. |
| RATn | LFO rate | Synth: LFO DEST | destination | RAT1 to RAT4: the rate of LFO 1 to 4, up to two octaves either way. |
| RC | Record | WAVE: REC | arm / stop | Two clicks clockwise within a second arm; one anticlockwise stops. |
| RCHN | Ratchet chance | ARP: VARY | 0–100 | Chance of a step ratcheting. |
| RDEC | Ratchet decay | ARP: PLAY | 0–100 | How much each ratchet repeat is quieter. |
| REC | Record mode | ARP: ACFG | NOTE / CHORD | How arp output is written when recording. |
| REL | Release | Insert effect: LMT | 50–500 ms | How fast the limiter recovers, in milliseconds. |
| REL | Release (CC 72) | MIDI track: SOUND | 0–127 | Sends the release-time controller. |
| REL | Release | Synth: F ENV, AMP; FM: OP1–OP6, F ENV; Macro, Wavetable, Wave seq: AMP, F ENV; Sampler: AMP | varies | Time the envelope takes to fall to silence after the note ends. |
| RELn | Envelope release | Synth, FM: LFO DEST | destination | FM, REL1 to REL6: the release of operator 1 to 6. Synth, REL2 to REL5: the release of an extra envelope (MENV). |
| RELS | Release | Insert effect: DUCK | 10–1000 ms | How fast the track comes back after ducking. |

| SHORT | FULL NAME | BELONGS TO | VALUES | WHAT IT DOES |
|---|---|---|---|---|
| RESB | Filter B resonance | Synth: FLT B | 0–100 | Resonance of the second filter. |
| RESO | Resonance (CC 71) | MIDI track: MIDI | 0–127 | Sends the resonance controller to the external instrument. |
| RESO | Resonance | Synth, FM, Macro, Wavetable, Wave seq, Sampler: FILTR; Insert effect: FLT, WAH | 0–100 | Emphasis at the cutoff frequency. The LADR type self-oscillates at full resonance. |
| RESO | Formant resonance | TRK FX: TALK | 0–100 | Sharpness of the vowel formants. |
| RHYT | Rhythm | ARP: ACFG | OFF / ON | Plays the arp in the tapped rhythm. |
| RING | Ring (particle noise model) | Macro: MACRO | 0–100 | Third macro of the DUST model, neutral at 50. Above: the particles ring at the pitch of the key. Below: a high-pass thins them into a hiss that follows the key. |
| RO.D | Rotary depth | TRK FX: MODFX | 0–100 | Amount of rotary speaker; 0 bypasses it. |
| ROLL | Roll | ARP: ROLL | OFF / ON | Hold-to-ratchet for every pad that plays a note. |
| ROOT | Root note | Sampler: SAMPL | 0–127 note | The key at which the sample plays at its original pitch; in KEYS slice play, the key that plays slice 1. |
| RO.S | Rotary speed | TRK FX: MODFX | 0–100 | Speed of the rotary speaker. |
| ROUT | Filter routing | Synth: FLT B | SNGL SER PAR | SNGL: filter A only. SER: A into B. PAR: A and B side by side. |
| RP | Repeats | SONG: bar | count | Times the row repeats. |
| RPT | Repeats | ARP: ARP | 1–8 | Times each note is repeated before moving on. |
| RPTS | Repeat limit | ARP: ROL2 | 0–16 | Stops after this many repeats; 0 = no limit. |
| RT | Root | SCALE: bar | C–B | Root note of the scale. |
| RTCH | Ratchet | ARP: PLAY | 1–4 | Repeats inside each arp step. |
| RTCH | Arp ratchet lock | TRK FX: ARPLK | 1–4 | Per-step lock of the arpeggiator's ratchet count. |
| RVB | Reverb send (CC 91) | MIDI track: SOUND | 0–127 | Sends the reverb-level controller. |
| RVRS | Reverse chance | Sampler: GRAIN | 0–100 | Proportion of grains played backwards. |
| RW | Row | SONG: bar | row | Selects the row (with fewer than two rows there is nothing to select, and the name bar says so). |
| S1–S16 | Step shape | Wave seq: SHP A / B | SIN SAW SQR TRI PLS NOIS | SHP A / SHP B pages: waveform of steps 1 to 16. |
| SAVE | Save project | SETTINGS: PROJ | four clicks | Saves to the current slot. |
| SAWS | Saws (virtual analog model) | Macro: MACRO | 0–100 | First macro of the VA model: number and blend of stacked saws. |
| SC | Scroll | WAVE: WAVE | position | Scrolls the zoomed view (zoom in with ZM first). |
| SCAN | Scan speed | Sampler: GRAIN | −2…2× | How fast the grain read position moves through the sample, independent of pitch. 0 freezes; negative runs backwards. |
| SCAT | Octave scatter | ARP: VARY | 0–100 | Chance of notes jumping octave. |
| SCRV | Kick sweep curve | Drums: KICK | −100…+100 | Kick machine: where in the sweep time the fall happens; negative = nearly all at the start. |
| SENS | Sensitivity | Insert effect: WAH | 0–100 | How far the playing level opens the wah (auto-wah). |
| SENS | Sensitivity | Sampler: SLICE | 0–100 | How readily a hit is detected in TRNS mode; higher finds quieter hits. |

| SHORT | FULL NAME | BELONGS TO | VALUES | WHAT IT DOES |
|---|---|---|---|---|
| SG | Song on | SONG: bar | OFF / ON | Arms song mode: Play then follows the rows. |
| SHAP | Kick shape | Drums: KICK | 0–100 | Kick machine: the body's waveform, from a sine through a triangle to a squared-off wave. |
| SHAP | Grain shape | Sampler: GRAIN | HANN TRI PERC SWEL | Grain window: HANN smooth, TRI, PERC percussive, SWEL swelling. |
| SHFT | Shift (formant model) | Macro: MACRO | 0–100 | Second macro of the FORM model: formant shift. |
| SHFT | Formant shift | TRK FX: TALK | −100…+100 | Moves all formants down or up: larger or smaller mouth. |
| SHIM | Shimmer amount | Insert effect: SHIM | 0–100 | How much pitch-shifted signal is fed into the tail. |
| SHP∗ | Shape, all steps | Wave seq: LANES | SIN SAW SQR TRI PLS NOIS | Sets every step's wave at once. |
| SHPE | LFO shape | FM, Wavetable, Wave seq, Sampler, Drums: LFO1 / LFO2 | SIN TRI SQR SAW RND SRND RAMP EXPU EXPD | LFO waveform: SIN TRI SQR SAW RND (stepped random) SRND (smooth random) RAMP EXPU EXPD. |
| SHPE | LFO shape | Insert effect: TRM, PAN | SIN TRI SQR | Waveform of the effect's own LFO. |
| SHPE | Shape (wavefolder model) | Macro: LFO1 / LFO2, MACRO | varies | Second macro of the FOLD model: input waveform. |
| SIZE | Room size | Insert effect: RVB, HALL | 0.25–2× | Size of the reverb space. |
| SIZE | Grain size | Sampler: GRAIN | 0.005–0.5 s | Length of each grain. |
| SIZE | Cycle / grain size | Sampler: STRCH | 0.005–0.5 s | Length of the pieces the stretch works in. Short is buzzy and tight, long is smooth and smeared. |
| SL | Slices | WAVE: WAVE | 1–64 | Number of slices (SLCS). |
| SLCS | Slices | Sampler: SLICE | 1–64 | GRID mode: number of equal slices. TRNS mode: shows how many hits were found; turning it down keeps only the strongest (1 = no limit, up to 64). |
| SLEN | Slice length | Sampler: SLICE | 0–100 | Shortens every slice to this share of its length, for gated, choppy playback. |
| SLEN | Shape lane length | Wave seq: LANES | 1–16 | Length of the wave-shape lane. |
| SLIC | Slice select | Sampler: SLICE | 0–100 | Which slice plays. |
| SLID | Slide | Wave seq: LANES | OFF ON | On: lane values glide between steps instead of jumping. |
| SLOP | Tuning slop | Synth: ANALG | 0–50 cents | Each new note starts slightly out of tune by up to this many cents. |
| SLOT | Project slot | SETTINGS: PROJ | P01–P32 | Which project slot LOAD, SAVE and SVAS act on. |
| SLOT | Insert slot | TRK FX: FX | slot number | Which of the track's insert slots the page shows. |
| SM | Sample select | WAVE: REC | list | Chooses the track's sample, or the USER wavetable's source. |
| SM | Key sample | WAVE: WAVE (wavetable track) | OFF, list | The sample this key is taken from (SMPn). Giving a key a sample selects the KEYS table. |
| SMP1–SMP8 | Key sample | Wavetable: K SMP | 19 values (App. C) | K SMP page: the sample key frame 1 to 8 is taken from. OFF: the key is not used. |
| SMPL | Sample | Sampler: SAMPL | 18 values (App. C) | Which sample the track plays: factory hits and loop, your files, the drum breaks (B080 … B170, named by tempo), and resampled takes. Choosing a break sets LEN to its length in bars. |
| SN | Snap | DRUMS: bar | 0–100 | SNAP of the selected drum pad. |

| SHORT | FULL NAME | BELONGS TO | VALUES | WHAT IT DOES |
|---|---|---|---|---|
| SNAP | Snap | Drums: DR | 0–100 | The front edge of the hit. 50 leaves the drum as it is; below, the onset is softened; above, the first milliseconds are lifted and the tonal drums get a click. TOM and KSYN set their attack time with it, CLAP also the length of each clap. |
| SND1 | Send 1 | TRACKS, TRK FX: SND1 / TRACK | 0–100 | Level sent to bus 1 (reverb by default). |
| SND2 | Send 2 | TRACKS, TRK FX: SND2 / TRACK | 0–100 | Level sent to bus 2 (delay by default). |
| SOFT | Soft start | Synth: ANALG | 0–50 cents | Pitch overshoot at the start of each note, in cents. |
| SP | Track speed | STEP: bar | 1/4 … 4x | Speed of the selected track. |
| SPCE | Spice | ARP: VARY / MUTA | 0–100 | Variation macro: how adventurous the variations are. |
| SPD | Speed | Insert effect: ROT | 0–100 | Rotary speaker speed, slow to fast. |
| SPD | Detector speed | Insert effect: TRNS | 0–100 | How quickly the transient shaper reacts. |
| SPD | Speed (bowed string model) | Macro: MACRO | 0–100 | Fourth macro of the BOW model: bow speed. |
| SPD | Stretch speed | Sampler: STRCH | 0.25–4× | Playback speed with the pitch kept (except TAPE): 0.25× to 4×. In SMER the speed is a sixteenth of the value shown. |
| SPD | Track speed | TRK FX, STEP: TRACK / bar | 1/4 1/2 1x 2x 4x | Playback speed of the track against the tempo. STEP shows it as SP. |
| SPD | Scan speed | Wavetable: SCAN | 0.25–4× | Divides the scan's pass time: 2× runs through twice as fast. |
| SPRD | Stereo spread | Insert effect: VIB, LCR | 0–100 | How far the effect is spread across left and right. |
| SPRD | Spread (virtual analog model) | Macro: MACRO | 0–100 | Second macro of the VA model: detune between the stacked saws. |
| SPRD | Grain spread | Sampler: GRAIN | 0–100 | Stereo spread of the grains. |
| SPRD | Stereo spread | Sampler: STRCH | 0–100 | Spreads the stretch's grains across left and right (GRAN and SMER). |
| SPRD | Unison spread | Synth: VOICE | 0–100 | Detune between the stacked unison voices. |
| SPRD | Spread | Wavetable: OSC | 0–100 | Detune and width of the wavetable's doubled voices. |
| SPRY | Spray | Sampler: GRAIN | 0–100 | Random scatter of grain positions and timing. |
| SR | Source | WAVE: REC | MSTR, T1 … | What is recorded: the master output or one track after its effects. |
| SRC | Sidechain source | Insert effect: DUCK | 17 values (App. C) | Which signal makes the track duck: SELF or another track T1 to T16. |
| SSIZ | Stretch size | Sampler: LFO DEST | destination | The SIZE cell of the STRCH page (SIZE alone is the grain size). |
| SSPR | Stretch spread | Sampler: LFO DEST | destination | The SPRD cell of the STRCH page (SPRD alone is the grain spread). |
| ST | Start | WAVE: WAVE | position | Start marker (the sampler's STRT). |
| STCK | Unison stack | Synth: VOICE | OFF 2 4 8 | Number of detuned copies stacked on every note (off, 2, 4, 8). |
| STEP | Scan steps | Wavetable: SCAN | 4 8 16 32 64 | Number of plateaus (STEP mode) or of new random frames per pass (RAND mode). |

| SHORT | FULL NAME | BELONGS TO | VALUES | WHAT IT DOES |
|---|---|---|---|---|
| STER | Stereo | Insert effect: TRM | 0–100 | Offsets left and right so the tremolo moves across the stereo field. |
| STGS | Stages | Insert effect: PHS | 2–6 | Number of phaser stages. |
| STGS | Chain stages | Macro, Wavetable, Wave seq: AMP | varies | Number of envelopes in the chain. |
| STRC | Stretch (harmonic model) | Macro: MACRO | 0–100 | Fourth macro of the HARM model: spreads the harmonics apart, as on a stiff string or a bar; the tone turns into a bell. |
| STRM | Strum | ARP: PLAY | OFF / ON | Plays held notes as a strummed chord. |
| STRP | Strip target | ARP: VARY | OFF SPCE DICE DENS SCAT RCHN GVAR GATE | Which macro the mod strip drives while this track's arp is on. |
| STRT | Start | Sampler: SAMPL | 0–100 | Where playback starts in the sample. |
| SUB | Sub level | Wavetable: OSC | 0–100 | Level of the sub oscillator one octave down. |
| SUS | Sustain gain | Insert effect: TRNS | −100…+100 | Boosts or reduces the tail after each hit. |
| SUS | Sustain | Synth: F ENV, AMP; FM: OP1–OP6, F ENV; Macro, Wavetable, Wave seq: AMP, F ENV; Sampler: AMP | 0–100 | Level held while the note is down. |
| SUSn | Envelope sustain | Synth, FM: LFO DEST | destination | FM, SUS1 to SUS6: the sustain of operator 1 to 6. Synth, SUS2 to SUS5: the sustain of an extra envelope (MENV). |
| SV | Save | PRESET: bar | four clicks | Saves the track's sound as a user preset (User 1, User 2, ...); the Mac then asks for a name. |
| SVAS | Save as | SETTINGS: PROJ | four clicks | Saves to the slot chosen with SLOT. |
| SWEP | Kick sweep | Drums: KICK | 0–72 semitones | Kick machine: how far above TUNE the pitch starts, in semitones. |
| SWNG | Swing | SETTINGS, ARP: SETUP / ARP / ROL2 | 0–100 | Delays every other step. SETUP: the pattern's swing. ARP and ROL2: the arp's and the roll's own. |
| SWPT | Kick sweep time | Drums: KICK | 0.002–1 s | Kick machine: how long the pitch takes to fall to TUNE. |
| SYNC | LFO tempo sync | FM, Macro, Wavetable, Sampler, Drums: LFO1 / LFO2 | OFF ON | On: RATE (or DIV) is a tempo division instead of Hz. |
| SYNC | Stretch tempo sync | Sampler: STRCH | OFF ON | On: the loop is stretched to last exactly LEN at the current tempo, and follows tempo changes. |
| SYNC | Tempo sync | Wave seq: LANES, LFO1 / LFO2 | OFF ON | On: step times follow the tempo. |
| SYNC | Scan sync | Wavetable: SCAN | OFF ON NAT | OFF: the pass lasts TIME. ON: it lasts LEN at the tempo. NAT: it lasts as long as the original recording (USER table). |
| T1–T16 | Step time | Wave seq: TIM A / B | 0.03–2 s | TIM A / TIM B pages: duration of steps 1 to 16. |
| TABL | Table | Wavetable: WAVE | 25 values (App. C) | Which wavetable: 23 built-in tables, USER (a sample cut into frames) or KEYS (a few waves picked out of samples, the frames between them computed). |
| TAP | Tap rhythm | ARP: ACFG | OFF / ON | While on, every pad press taps a step of a custom rhythm. |

| SHORT | FULL NAME | BELONGS TO | VALUES | WHAT IT DOES |
|---|---|---|---|---|
| TAPS | Taps | Insert effect: MTAP | 1–8 | Number of echoes in the multi-tap pattern. |
| TB | Table | WAVE: WAVE (wavetable track) | list | The track's table (TABL). |
| TENS | Tension | Insert effect: SPRG | 0–100 | Spring tension: tighter and brighter, or looser and boingier. |
| TH | Threshold | WAVE: REC | level | Recording starts when the source passes this level. |
| THR | Threshold | Insert effect: CMP | −60…0 dB | Level above which the compressor works. |
| THR | Master compressor threshold | SETTINGS: MASTR | −60–0 dB | Level above which the master compressor works. |
| THRS | Threshold | Insert effect: DUCK | −60…0 dB | Level of the source above which the track ducks. |
| TILT | Tilt (harmonic model) | Macro: MACRO | 0–100 | First macro of the HARM model: spectral tilt. |
| TILT | Key tilt | Wavetable: KEYS | −600–600 | Makes every key of the KEYS table darker (down) or brighter (up), by up to 6 dB per octave. |
| TIME | Strum time | ARP: PLAY | 0–100 ms | Gap between strummed notes. |
| TIME | Delay time | Insert effect: DLY, RVSE, TAPE, LCR, MTAP | varies | Time between echoes. With SYNC on it picks a tempo division. |
| TIME | Scan time | Wavetable: SCAN | 0.05–60 s | Time one pass through the table takes, while SYNC is off. |
| TLEN | Time lane length | Wave seq: LANES | 1–16 | Length of the step-duration lane. |
| TN | Tune | DRUMS: bar | 20 Hz–12 kHz | TUNE of the selected drum pad. |
| TO | Tone | DRUMS: bar | 0–100 | TONE of the selected drum pad. |
| TONE | Tone | Drums: DR | 0–100 | Brightness, in the way of each type: the click of FMKK, how far KICK's pitch falls, MKIK's resonance, the noise band of the snares, the width of CLAP's band, the high-pass of the hats and cymbal, TOM's stick noise, RIM's noise, a broad hiss to a pitched rattle on SHKR, CLAV's upper partial, sine to square on SUB, a low-pass on KSYN. |
| TONE | Tone | Insert effect: DRV, WSH, AMP | varies | Brightness of the effect. |
| TR | Transpose | CHORD: bar | −24 to +24 | Moves every chord pad by semitones; what is stored stays as played. |
| TR | Transpose | SONG: bar | semitones | Transposes pitched tracks for this row. |
| TRIG | LFO trigger mode | FM, Macro, Wavetable, Wave seq, Sampler, Drums: LFO1 / LFO2 | FREE KEY ONE HALF HOLD | FREE runs on; KEY restarts on each note; ONE plays one cycle then holds; HALF plays half a cycle; HOLD samples one value per note. |
| TRKB | Filter B key tracking | Synth: FLT B | 0–100 | How far filter B's cutoff follows the note. |
| TUN2 | Oscillator 2 tune | Synth: OSC | −24…24 semitones | Pitch of oscillator 2 relative to oscillator 1, in semitones. |
| TUN3 | Oscillator 3 tune | Synth: OSC3 | −24…24 semitones | Pitch of oscillator 3 in semitones. |
| TUNE | Tune | Drums: DR | 20 Hz–12 kHz | Pitch of the drum voice in Hz. |
| TUNE | Tune | Synth: OSC3; FM: RATIO; Macro: MACRO; Wavetable: OSC; Wave seq: SEQ | varies | Coarse pitch of the whole sound in semitones. |
| TUNE | Feedback tune | TRK FX: DIRT | 30 Hz–4 kHz | Frequency the DIRT block's feedback path resonates at. |

| SHORT | FULL NAME | BELONGS TO | VALUES | WHAT IT DOES |
|---|---|---|---|---|
| TY | Type | DRUMS: bar | 14 models | TYPE of the selected drum pad. |
| TYPB | Filter B type | Synth: FLT B | SVF SEM LADR DIOD COMB OFF | Model of the second filter. |
| TYPE | Drum type | Drums: DR | 15 values (App. C) | Synthesis model of this drum voice (15 models; Appendix C). KSYN, the kick machine, adds the KICK page. |
| TYPE | Filter type | Synth, FM, Macro, Wavetable, Wave seq, Sampler: FILTR; Insert effect: FLT | varies | Filter model: SVF, SEM, LADR (ladder), DIOD (diode), COMB, or OFF. |
| TYPE | Distortion type | TRK FX: DIRT | SOFT HARD FOLD ASYM | SOFT, HARD, FOLD (wavefolder) or ASYM (asymmetric). |
| TYPE | Insert effect type | TRK FX: FX | 30 types | Effect in the selected slot. Changing it replaces the effect and its four values. |
| VAR | Variation | ARP: VARY | number | Selects a stored variation. |
| VDLY | Vibrato delay (CC 78) | MIDI track: SOUND | 0–127 | Sends the vibrato-delay controller. |
| VDPT | Vibrato depth (CC 77) | MIDI track: SOUND | 0–127 | Sends the vibrato-depth controller. |
| VEL | Strum velocity | ARP: PLAY | 0–100 | Velocity change across the strum. |
| VEL | Step velocity | STEP: TRIG | 0–127 | Velocity of the step's notes. |
| VEL | Velocity to cutoff | Synth, FM, Macro, Wavetable, Wave seq: F ENV | 0–100 | Harder notes open the filter further. |
| VEL | Velocity sensitivity | Synth, Wavetable, Wave seq, Sampler: AMP | 0–100 | 0: every note at full level. 100: level follows how hard you play. |
| VEL | Arp velocity lock | TRK FX: ARPLK | 0–100 | Per-step lock of the arpeggiator's velocity. |
| VELF | Velocity to filter | Sampler: FILTR | 0–100 | Harder notes open the filter further. |
| VELP | Velocity to position | Wavetable: WAVE | 0–100 | Harder notes start further into the table. |
| VIBR | Vibrato | Wavetable: OSC | 0–100 | Depth of pitch vibrato. |
| VL | Velocity | CHORD: bar | as played / pad | Each note at the level it was stored with, scaled by the pad; or every note at the pad's velocity. |
| VL | Volume | PRESET: bar | 0–100 | Track volume, to level-match while browsing. |
| VOIC | Voices | Insert effect: CHR | 1–3 | Number of chorus voices. |
| VOL | Volume (CC 7) | MIDI track: MIDI | 0–127 | Sends the volume controller. |
| VOL | Track volume | TRACKS: VOL | 0–100 | Mixer page: knob N is the volume of track N. |
| VOWL | Vowel (formant model) | Macro: MACRO | 0–100 | First macro of the FORM model: vowel. |
| VOWL | Vowel | TRK FX: TALK | 0–100 | Position through the vowels A, E, I, O, U. |
| VOX | Voices | Synth: VOICE; Sampler: AMP | varies | Maximum number of notes that sound at once. |
| VRAT | Vibrato rate (CC 76) | FM, Macro, Wavetable, Wave seq, Sampler, Drums: STRIP; MIDI track: SOUND | varies | Sends the vibrato-rate controller. |
| WAV1 | Oscillator 1 kind | Synth: OSC | 10 values (App. C) | What oscillator 1 plays: an analog wave (SIN SAW SQR TRI PLS NOIS) or a wavetable (WT), FM pair (FM), formant (FORM) or harmonic (HARM) oscillator. |
| WAV2 | Oscillator 2 kind | Synth: OSC | 10 values (App. C) | What oscillator 2 plays; the same choices as oscillator 1. |

| SHORT | FULL NAME | BELONGS TO | VALUES | WHAT IT DOES |
|---|---|---|---|---|
| WAV3 | Oscillator 3 kind | Synth: OSC3 | 10 values (App. C) | What oscillator 3 (the sub oscillator) plays; the same choices as oscillator 1. |
| WDTH | Width | Insert effect: ENS | 0–100 | Stereo width of the ensemble. |
| WET | Insert wet level | TRK FX: FX | 0–100 | How much of the effect is heard; 0 takes it out. |
| WHL | Mod strip to cutoff | FM, Macro, Wavetable, Wave seq: F ENV | −100…+100 | The mod strip opens (or, negative, closes) the filter. |
| WRAP | Wrap | Wavetable: KEYS | OFF ON | On: after the last key the KEYS table returns to the first, so a looped scan has no jump. |
| WSHP | Waveshape variation | Synth: ANALG | 0–100 | Random change of the waveform on every note: a component tolerance at low settings, a difference heard from note to note towards the top. |
| WVIB | Strip vibrato | FM, Macro, Wavetable, Wave seq, Sampler, Drums: STRIP | 0–1 semitones | STRIP page: vibrato the mod strip brings in, in semitones at the top of the strip. 0 = none. |
| XFAD | Crossfade | Sampler: STRCH | 0–100 | Overlap between successive pieces of the stretch; in SLCE, how quickly a slice is faded when the next arrives. |
| XFAD | Crossfade | Wave seq: SEQ | 0–100 | How much each step's wave blends into the next. |
| XMOD | Cross-modulation | Synth: OSC | 0–100 | How strongly oscillator 2 frequency-modulates oscillator 1: through-zero linear FM, so the note stays in tune and whole-number tunings of oscillator 2 (TUN2 at 0, 12, 19, 24) give harmonic tones at any depth; other tunings give bells and clangs. |
| ZM | Zoom | WAVE: WAVE | factor | Zooms the waveform view. |

# Appendix B Gestures at a glance

| TO... | DO THIS |
|---|---|
| Go to a screen | `Shift` + pad |
| Next / previous knob page | Knob-page buttons |
| Jump to a page | `Func` + top-row pad |
| Select a track | Hold ▲ or ▼ + top pad N; or `Shift` + ▲ / ▼; or top pad on `TRACKS` |
| Mute / solo a track | Hold ▲ or ▼ + bottom pad N / `Func` + bottom pad on `TRACKS` |
| Move a control on all tracks | Hold ▲ or ▼ + turn |
| See what Shift or Func does here | Hold `Shift` or `Func` alone; knob-page buttons turn the cards |
| Fine / coarse | `Shift` + turn / `Func` + turn |
| Add or remove a step | Tap a pad on `STEP` |
| Set a step's note | Hold the step + key |
| Lock a knob to a step | Hold the step + turn |
| Remove that lock | Hold the step + `>` + turn |
| Change pattern | Hold `>` + pad |
| Keep / reload the pattern | `>` + `Rec` / `>` + `Play` |
| Copy / clear / paste | `Func` + `Rec` / `Play` / `>` (a whole pattern: hold half a second) |
| Undo / redo | `Func` + `Shift` + `Play` / `Rec` |
| Record | Hold `Rec` + `Play`; again = free / quantized; `Play` stops and shows the steps |
| Fill | Hold `Shift` + `Rec` |
| Stop, rewind and silence everything | `Shift` + `Play` |
| Tap tempo | `Func` + `Settings`, tapped |
| Arp on / off | `Arp` (opens the screen), `Arp` again |
| Latch | `Shift` + `Arp` |
| Scale lock / scale screen | `Scale` / `Shift` + `Scale` |
| Pressure range on a knob | Turn it, then `Func` + ▲ (high end) or ▼ (low end); again clears |
| Pin the playing pads | On `DRUMS` or `SCALE` : `Shift` + hold that screen's bottom pad half a second |
| Confirm something destructive | Four clicks clockwise on its knob |
| Split the keyboard | `Shift` + `Oct−` / `Oct+` |

# Appendix C Value lists

| LIST | VALUES, IN ORDER |
|---|---|
| Engines ( `ENG` ) | Synth · FM · Macro oscillator · Wavetable · Wave sequencer · Sampler · Drums · MIDI track |
| Oscillator waves | SIN SAW SQR TRI PLS NOIS WT FM FORM HARM |
| Filter types | SVF SEM LADR DIOD COMB OFF |
| LFO shapes ( `SHPE` ) | SIN TRI SQR SAW RND SRND RAMP EXPU EXPD |
| LFO trigger modes ( `TRIG` ) | FREE KEY ONE HALF HOLD |
| LFO divisions ( `DIV` ) | 8BAR 4BAR 2BAR 1BAR 1/2D 1/2 1/4D 1/2T 1/4 1/8D 1/4T 1/8 1/16D 1/8T 1/16 1/16T 1/32 (D = dotted, T = triplet) |
| Effect sync divisions | 1/128 1/64 1/32T 1/32 1/16T 1/32D 1/16 1/8T 1/16D 1/8 1/4T 1/8D 1/4 1/2T 1/4D 1/2 1/2D 1BAR 2BAR 4BAR 8BAR 16BAR |
| Insert effect types | CHR DRV FLG PHS TRM DLY RVB EQ CMP LMT FLT ENS PAN ROT WSH RED HALL SPRG SHIM RVSE TRNS RING VIB WAH AMP GEQ TAPE LCR MTAP DUCK |
| Drum types | FMKK KICK MKIK SNAR FMSN CLAP HAT OHAT CYMB TOM RIM SHKR CLAV SUB KSYN |
| Sampler play modes | FWD REV 1SHT GATE LOOP PING GRAN |
| Slice play modes | ONE KEYS NEXT RAND |
| Stretch algorithms | OFF TAPE CYCL GRAN SLCE SMER WSOL |
| Stretch lengths (bars) | 1/16 1/8 1/4 1/2 1 2 4 8 16 32 |
| Wavetable scan modes | OFF RUN LOOP PING STEP RAND |
| Wavetable frame options | 64 128 256 A64 A128 A256 |
| Grain shapes | HANN TRI PERC SWEL |
| Wavetables | SAW SQR TRI SIN PWM FORM HARM SPEC DIGI USER KEYS VOWL SYNC FOLD FM1 FM3 BELL ORGN NOIS RESO WLSH PRIM OCTV PDST COMB |
| Macro models | VA FM FOLD FORM HARM PLCK DUST BOW |
| FM algorithms | 6>1 2+3 3x2 2x3 1>3 ADD 2+4 4+2 |
| Trig conditions | ALL PROB A:B FILL !FIL 1ST !1ST PRE !PRE |
| Arp modes | Up; Down; Up-down; Random; As played; Chord repeat; Walk; Play order; Converge; Diverge; Poly arp (all notes, accented); Random with octave jumps; Parallel (PARA: every key runs its own climb through the octaves from the moment it is pressed, side by side with the others; keys pressed together climb together); Pedal (PEDL: the lowest note sounds before each higher one in turn – 1 2 1 3 1 4); Rising pedal (PDL+: the same, and after each pass the next note up becomes the pedal – 1 2 1 3 1 4, 2 3 2 4, 3 4) |
| Arp rates | 1/1 1/2 1/4 1/4T 1/8 1/8T 1/16 1/16T 1/32 1/32T 1/64 1/64T (the knob runs slowest to fastest) |
| Roll divisions | 1/2 1/4 1/4T 1/8 1/8T 1/16 1/16T 1/32 1/32T 1/64 1/64T |
| Track speeds | 1/4 1/2 1x 2x 4x |

| LIST | VALUES, IN ORDER |
|---|---|
| Scale pages (built in) | Chrom Major Minor Penta Blues Whole Dorian Mixo HrMin mPent Phryg Lydia |
| Preset categories | ALL INIT BASS PAD LEAD KEYS FX PERC KIT |
| Clock sources | INT EXT AUTO LINK |

# Appendix D Glossary of terms

| TERM | MEANING |
|---|---|
| Bar screen | A screen with a picture or list and a strip of eight two-letter knob labels along the bottom. |
| Bus | One of two shared effects that every track can send to. |
| Cell | One of the eight positions on the display; the knob in the same position edits it. |
| Condition | A rule on a step that decides whether it plays on a given pass. |
| Engine | The sound generator of a track. |
| Fill | A held state in which steps marked FILL play. |
| Four clicks | The confirmation for anything destructive: four clockwise clicks on its knob. |
| Grain | A short slice of a sample, played many at a time in granular mode. |
| Grid screen | A screen with eight cells in two rows of four. |
| Insert | An effect in a slot on one track. |
| Knob page | One set of eight cells; the knob-page buttons move between them, going round at the ends. |
| Latch | Notes or an arp carrying on after the keys are released. |
| Lock (parameter lock) | A knob value that applies to one step only. |
| Micro-timing | A step's offset from the grid, up to half a step either way. |
| Name bar | The top line of the display. |
| Page key | Func + a top-row pad: jumps straight to a page. |
| Pattern | One of 16 sets of steps for all tracks. |
| Pin | Keeping the DRUMS or SCALE pads playable on every screen. |
| Pressure | Polyphonic aftertouch: how hard each pad is pressed after it is struck, per pad. |
| Pressure range | A range on a knob that a bottom-row pad sweeps with pressure. |
| Ratchet | Several quick repeats inside one step. |
| Resampling | Recording Neptune's own output into a new sample. |
| Roll | A pad repeating while held, faster with pressure. |
| Route | A pressure destination and depth on a track. |
| Row | One line of the song: a pattern, a repeat count and a transposition. |
| Send | The amount of a track fed to a bus. |
| Slice | One piece of a sample, cut evenly or at a hit. |
| Time stretch | Changing how long a sound lasts without changing its pitch. |
| Transient | The sudden start of a sound, such as a drum hit; where transient slicing cuts. |
| Step | One position in a track's pattern. |

| TERM | MEANING |
| --- | --- |
| Track layer | What is available while ▲ or ▼ is held: track select, mute, and all-track turns. |
| Trig | A step that is switched on. |